Immediate PAH therapy should be considered in patients who present with intermediate-/high-risk PAH at diagnosis	IIa	C
Patients with low-risk PAH should be re-evaluated 3–4 months after discontinuing the suspected drug or toxin, and PAH therapy may be considered when the haemodynamics have not normalized	IIb	C

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PAH, pulmonary arterial hypertension; PH, pulmonary hypertension.
[a]Class of recommendation.
[b]Level of evidence.

7.2. Pulmonary arterial hypertension associated with connective tissue disease

Pulmonary arterial hypertension is a well-known pulmonary vascular complication of SSc,[173,502–504] systemic lupus erythematosus (SLE),[505–507] mixed CTD,[506] and, rarely, dermatomyositis[508] and Sjögren's syndrome.[509] Conversely, the relationship between rheumatoid arthritis and PAH is not established.[510] After IPAH, PAH-CTD is the second most prevalent type of PAH in western countries.[511]

Systemic sclerosis, particularly in its limited variant, represents the main cause of PAH-CTD in Europe and the USA (SLE being more common in Asia).[173,502,506] The prevalence of pre-capillary PH in large cohorts of patients with SSc is 5–19%.[173,502] In these patients, PH may occur in association with ILD[504,512] or as a result of PAH,[173,502–504,506] sometimes with features of venous/capillary involvement.[504,513] Moreover, group 2 PH-LHD is also common due to myocardial SSc involvement.[504,514] Of note, patients with SLE may also present with PAH, LHD, ILD, and CTEPH (mostly in the setting of antiphospholipid syndrome). It is therefore essential to carefully determine which mechanism is operative in a given patient, since this will dictate treatment in the context of a multifaceted disease.

Cluster analysis performed in patients with SSc has shown that pre-capillary PH can be characterized into distinct clusters that differ in prognosis.[503] One cluster, characterized by the presence of extensive ILD, and another by severely impaired haemodynamics carried a dismal prognosis, while the two others showed either the absence of ILD or the presence of limited ILD, with mild-to-moderate risk PAH and a relatively favourable overall prognosis.[503]

7.2.1. Epidemiology and diagnosis

There is a strong female predominance in PAH-CTD (female/male ratio 4:1), and mean age at diagnosis is commonly >50 years, especially in SSc.[173,502–511,513,515,516] In the setting of a CTD, patients may present with concomitant disorders such as ILD, and have shorter survival compared with patients with IPAH.[503] The unadjusted risk of death for PAH-SSc compared with IPAH is 2.9, and the predictors of outcome are broadly similar to those for IPAH.[516,517] Symptoms and clinical presentation are also similar to IPAH, and some patients thought to have IPAH can be identified as having an associated CTD by careful clinical examination and immunological screening tests. Chest CT is recommended for evaluating the presence of associated ILD or PVOD/PCH.[504,513,515] An isolated reduction of DLCO is common in PAH-CTD.[173,502–504]

Resting echocardiography combined with other tests is recommended as a screening test in asymptomatic patients with SSc, followed by annual assessments. Screening/early detection is discussed in *Section 5.3.1*. In other CTDs, PH screening in the absence of suggestive symptoms is not recommended, while echocardiography should be performed in the presence of symptoms. As in other forms of PAH, RHC is recommended in all cases of suspected PAH-CTD to confirm diagnosis, determine severity, and rule out LHD.[504]

7.2.2. Therapy

Drugs for PAH should be prescribed in PAH-CTD according to the same treatment algorithm as in IPAH (*Figure 9*). Patients with PAH-CTD have been included in most of the major RCTs for regulatory approval of PAH therapy.[518] Some aspects of PAH-CTD treatment differ according to the associated CTD.[506] Immunosuppressive therapy combining glucocorticosteroids and cyclophosphamide may result in clinical improvement in patients with SLE- or mixed CTD-associated PAH,[506] while it is not recommended in PAH-SSc.[519] Patients with SSc and other CTDs may have ILD and/or HFpEF, which needs to be considered when initiating PAH therapy.[504,515] In SSc, the long-term risk/benefit ratio of oral anticoagulation is unfavourable because of an increased risk of bleeding, while VKAs are recommended in PAH-CTD with a thrombophilic predisposition (e.g. antiphospholipid syndrome).[319]

Subgroup analyses of patients with PAH-SSc enrolled in RCTs performed with monotherapy or combination therapy of ERAs, PDE5is, sGC stimulators, prostacyclin receptor agonists, epoprostenol, and prostacyclin analogues have shown positive effects vs. placebo.[301,401,519,520] In some of these trials, the magnitude of the response in the PAH-CTD subgroup was lower than in the IPAH subgroup.[519,520] Continuous i.v. epoprostenol therapy improved exercise capacity, symptoms, and haemodynamics in a 3-month RCT in PAH-SSc.[401] However, a retrospective analysis showed a better effect of i.v. epoprostenol on survival in IPAH compared with PAH-SSc.[521] The choice of PAH therapy in the context of SSc and its systemic manifestations may consider other vascular damage such as digital ulcers.[522]

Connective tissue disease should not be considered as an a priori contraindication for LTx.[523] This has been extensively studied in SSc, where a multidisciplinary approach optimizing SSc management before, during, and after surgery is recommended.[523] Indications and contraindications for transplantation have to be adapted to the specificities of CTD, with a special focus on digestive (gastro-oesophageal reflux disease and intestinal disease), cardiac, renal, and cutaneous involvement.[523]

Recommendation Table 15 — Recommendations for pulmonary arterial hypertension associated with connective tissue disease

Recommendations	Class[a]	Level[b]
In patients with PAH associated with CTD, treatment of the underlying condition according to current guidelines is recommended[166,167,419,524]	I	A

Continued

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In patients with PAH associated with CTD, the same treatment algorithm as for patients with IPAH is recommended	I	C

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PAH-CTD, pulmonary arterial hypertension associated with connective tissue disease; IPAH, idiopathic pulmonary arterial hypertension.
[a]Class of recommendation.
[b]Level of evidence.

7.3. Pulmonary arterial hypertension associated with human immunodeficiency virus infection

The use of highly active antiretroviral therapy (HAART), and advances in managing opportunistic infections have contributed to increased life expectancy in patients with HIV.[525,526] Consequently, the spectrum of complications has shifted towards other long-term conditions, including PAH. Clinical and histopathological findings in PAH associated with HIV infection (PAH-HIV) share many similarities with IPAH.[1,527] With the availability of HAART given in combination with PAH therapies, the prognosis of PAH-HIV has markedly improved in recent years.[526,528] In addition, the incidence of PAH-HIV has declined in parallel with the increasing availability of HAART.[528] Taken together, these effects on survival and incidence have resulted in a stable PAH prevalence in patients with HIV over recent decades. A French population study indicated that the prevalence of PAH in individuals with HIV infection was 0.46%, which is very similar to the prevalence before the HAART era.[177]

The pathogenesis of PAH-HIV remains unclear. There is no evidence of a direct role of HIV in the pathogenesis of PAH and, although present in inflammatory cells in the lungs, the virus itself has never been found in pulmonary vascular lesions of patients with PAH-HIV.[529] This suggests that an indirect action of viral infection on inflammation and growth factors may act as a trigger in a predisposed patient.

7.3.1. Diagnosis

Pulmonary arterial hypertension associated with HIV shares a clinical presentation with IPAH. Before the availability of HAART most patients were in WHO-FC III or IV at diagnosis. Nowadays, patients are diagnosed with much less severe symptoms and haemodynamics. Patients may present with other risk factors for PAH such as liver disease (chronic viral hepatitis B or C) or exposure to drugs or toxins. Patients with PAH-HIV are more likely to be male and i.v. drug abusers.[403,526] There is no correlation between the severity of PAH and the stage of HIV infection or the degree of immunodeficiency.[403,530] Because of its low prevalence, asymptomatic patients with HIV should not be screened for PAH. However, echocardiography should be performed in patients with unexplained dyspnoea to detect HIV-related cardiovascular complications such as myocarditis, cardiomyopathy, or PAH. Right heart catheterization is mandatory to confirm the diagnosis of PAH-HIV and to rule out LHD.[527]

Pulmonary arterial hypertension is an independent risk factor for death in patients with HIV. In the 1990s, before the availability of HAART, patients with PAH-HIV had poor outcomes, with a 3 year survival of <50%.[403] The overall survival has now improved and patients with PAH-HIV have a better prognosis than most patients with other forms of PAH.[526]

7.3.2. Therapy

Current recommendations for the treatment of PAH-HIV are largely based on data from IPAH.[25,26]

Treatment of PAH-HIV with HAART has improved functional status and survival in some retrospective studies.[525,526,531] The use of HAART in PAH-HIV is therefore recommended, irrespective of viral load and CD4+ cell count.

Anticoagulation is not recommended because of an increased risk of bleeding and drug interactions.[319,527] Patients with PAH-HIV are usually non-responders to acute vasoreactivity testing and therefore should not receive CCBs.[378]

The prospective, open-label, BREATHE-4 study showed that bosentan markedly improved WHO-FC, exercise capacity, quality of life, and haemodynamics after 16 weeks in patients with PAH-HIV.[532] In a long-term, retrospective series, bosentan therapy was associated with haemodynamic normalization in 10/59 patients.[533] Bosentan potentially interacts with antiretroviral drugs, and close monitoring is required when combined with HAART. Very few patients with PAH-HIV have been included in RCTs with ambrisentan and macitentan, and no definite conclusion can be drawn from those studies.

Positive effects of sildenafil and tadalafil in PAH-HIV have been established in case studies.[534,535] Interactions have been reported between PDE5is and protease inhibitors, resulting in major increases in PDE5i concentrations; these drugs should be introduced at low dosages with careful monitoring of potential side effects, including hypotension.[536,537] There are no data on the use of the sGC stimulator riociguat in PAH-HIV.

Treatment with i.v. epoprostenol resulted in significant improvement in WHO-FC, exercise capacity, haemodynamics, and survival in selected patients with PAH-HIV.[403,538] There are very few data on the use of i.v. or s.c. treprostinil or inhaled iloprost in PAH-HIV.[539,540]

There are no clinical trial data on the use of combination therapy for PAH-HIV. Given the lack of supporting evidence and potential safety concerns when PAH drugs are co-administered with antiretroviral drugs, initial monotherapy with PAH medication is recommended, followed by an individualized use of combination therapy in patients who do not reach a low-risk profile.

Recommendation Table 16 — Recommendations for pulmonary arterial hypertension associated with human immunodeficiency virus infection

Recommendations	Class[a]	Level[b]
In patients with PAH associated with HIV infection, antiretroviral treatment according to current guidelines is recommended[541,542]	I	A
In patients with PAH associated with HIV infection, initial monotherapy should be considered, followed by sequential combination if necessary, taking into consideration comorbidities and drug–drug interactions	IIa	C

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HIV, Human immunodeficiency virus; PAH, pulmonary arterial hypertension.
[a]Class of recommendation.
[b]Level of evidence.

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7.4. Pulmonary arterial hypertension associated with portal hypertension

Pulmonary arterial hypertension associated with portal hypertension, commonly referred to as PoPH, develops in 2–6% of patients with portal hypertension, with or without liver disease. In PAH registries, PoPH represents 5–15% of the patients.[543–545] Rarely, some patients with PoPH have portosystemic shunts in the absence of portal hypertension (congenital extrahepatic cavoportal shunts).[546] However, PoPH is distinct from hepatopulmonary syndrome (HPS), which is characterized by intrapulmonary vascular dilatations and hypoxaemia. Of note, HPS and PoPH can occur sequentially or concurrently in patients with portal hypertension.[547]

7.4.1. Diagnosis

The diagnosis of PoPH is based on the presence of otherwise unexplained pre-capillary PH in patients with portal hypertension or a portosystemic shunt. The diagnostic approach is the same as in other patients with suspected or newly detected PH. Transthoracic echocardiography is usually the first non-invasive assessment in patients with suspected PH, and echocardiography is also recommended as a screening tool in patients evaluated for liver transplantation. As patients with liver disease often have an elevated CO, TRV tends to overestimate PAP in these patients. Hence, RHC with comprehensive haemodynamic assessment is essential to confirm the diagnosis of PH and to distinguish PAH (with elevated PVR) from unclassified PH (with a normal PVR).

7.4.2. Therapy

Patients with unclassified PH (i.e. mPAP >20 mmHg, elevated CO, and PVR ≤2.0 WU) should be regularly followed-up but should not be treated with drugs approved for PAH.

In patients with an established diagnosis of PoPH, treatment should follow the same general principles as in other patients with PAH, taking into account the severity of underlying liver disease, the indication for liver transplantation, and the potential effects of PAH medication on gas exchange, which may deteriorate with vasodilators in patients with PoPH.[548,549] All drugs approved for PAH can principally be used to treat patients with PoPH, bearing in mind that these patients are usually excluded from registration studies. Nevertheless, various case series support the use of approved PAH medication in patients with PoPH. The largest series published so far reported on 574 patients with PoPH treated with various PAH drugs, mostly PDE5is or ERAs, alone and in combination.[545] Most patients (56.8%) were in Child–Pugh class A at the time of PAH diagnosis. At the first follow-up, which took place 4.5 months after starting treatment, improvements were seen in haemodynamics, WHO-FC, 6MWD, and BNP/NT-proBNP; survival at 5 years was 51%. In patients presenting with mild liver disease, the main causes of death were PAH progression and malignancy, whereas complications of liver disease were the most common causes of death in patients with advanced liver disease. The 5 year survival of patients who underwent liver transplantation ($n = 63$) was 81%.

The only RCT dedicated to the treatment of PoPH was PORTICO, a 12 week study that randomized 85 patients to macitentan ($n = 43$) or placebo ($n = 42$).[168] PORTICO met its primary endpoint, demonstrating a significant reduction in PVR from baseline (ratio of geometric mean 0.65; 95% CI, 0.59–0.72; $P < 0.0001$). There were, however, no differences between the two treatment groups in secondary outcome measures, including WHO-FC, 6MWD, and NT-proBNP.

7.4.2.1. Liver transplantation

Porto-pulmonary hypertension is not per se an indication for liver transplantation. Pulmonary arterial hypertension poses a major threat to patients who undergo liver transplantation when indicated for the severity of liver disease. In a historical series from the Mayo Clinic, severe PAH with mPAP ≥50 mmHg was associated with a 100% perioperative mortality rate. In patients with mPAP 35–50 mmHg and PVR >3.0 WU, mortality was still 50%.[550] In liver transplantation candidates with PAH, targeted medical therapy successfully improves haemodynamics and establishes eligibility for transplantation.[545,551–554] However, haemodynamic criteria for successful liver transplantation have not been firmly established. The International Liver Transplant Society proposed haemodynamic targets of mPAP <35 mmHg and PVR <5 WU, or mPAP ≥35 mmHg and PVR <3 WU in patients receiving PAH therapy, while acknowledging that these criteria need to be further validated.[175] An mPAP ≥45 mmHg is regarded as an absolute contraindication to liver transplantation.[175]

In patients with PoPH who successfully underwent liver transplantation, de-escalation or discontinuation of PAH medication is often feasible, but this has to be performed on an individual basis.[551,554]

Recommendation Table 17 — Recommendations for pulmonary arterial hypertension associated with portal hypertension

Recommendations	Class[a]	Level[b]
Echocardiography is recommended in patients with liver disease or portal hypertension with signs or symptoms suggestive of PH, and as a screening tool in patients evaluated for liver transplantation or transjugular portosystemic shunt	I	C
It is recommended that patients with PAH associated with portal hypertension are referred to centres with expertise in managing both conditions	I	C
In patients with PAH associated with portal hypertension, initial monotherapy should be considered, followed by sequential combination if necessary, taking into consideration the underlying liver disease and indication for liver transplantation	IIa	C
Liver transplantation should be considered on an individual basis in patients with PAH associated with portal hypertension, as long as PVR is normal or near normal with PAH therapy	IIa	C
Drugs approved for PAH are not recommended for patients with portal hypertension and unclassified PH (i.e. elevated mPAP, high CO, and a normal PVR)	III	C

mPAP, mean pulmonary arterial pressure; CO, cardiac output; PAH, pulmonary arterial hypertension; PH, pulmonary hypertension; PVR, pulmonary vascular resistance.
[a]Class of recommendation.
[b]Level of evidence.

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7.5. Pulmonary arterial hypertension associated with adult congenital heart disease

The presence of PH in adults with CHD has a negative impact on the natural course of CHD, and worsens clinical status and overall outcome.[555] Pulmonary arterial hypertension associated with adult CHD is included in group 1 of the PH clinical classification (*Table 6*) and represents a heterogeneous patient population. Post-capillary PH in adult CHD (e.g. systolic or diastolic, systemic, ventricular dysfunction in combination with shunt lesions or complex adult CHD, and systemic atrioventricular valve dysfunction) should be excluded to determine further management. A specific clinical classification (*Table 21*) is provided to better characterize PAH associated with adult CHD. Some complex CHDs are associated with congenital abnormalities of the pulmonary vascular tree leading to segmental PH. In segmental PH, one or more, but not all, segments of the lung(s) are hypertensive and each hypertensive area may present with PH of different severity, while other parts of the lung vasculature may be hypoplastic. Pulmonary atresia with ventricular septal defect and systemic-to-pulmonary collaterals is the most frequent condition, but other complex CHDs may also lead to segmental PH.

Approximately 3–7% of patients with adult CHD will eventually develop PAH; it is more frequently encountered in females, and the incidence depends on the underlying lesion and increases with age and age at defect closure.[556] The estimated prevalence of PAH in patients after correcting a simple cardiac defect is 3%.[557] The epidemiology of PAH associated with adult CHD is expected to change due to advances in diagnostic and therapeutic paediatric cardiology, resulting in fewer patients with simple adult CHD and more patients with complex lesions and/or closed defects who develop PAH in adulthood.[558]

The clinical presentation of Eisenmenger syndrome, an advanced form of adult CHD-associated PAH, is characterized by the multiorgan effects of chronic hypoxaemia, including cyanosis, and haematological changes, including secondary erythrocytosis and thrombocytopenia; the main symptoms are dyspnoea, fatigue, and syncope. Eisenmenger syndrome may also present with haemoptysis, chest pain, cerebrovascular accidents, brain abscesses, coagulation abnormalities, and sudden death. Patients with adult CHD and Down syndrome are at an increased risk of developing Eisenmenger syndrome.

Table 21 Clinical classification of pulmonary arterial hypertension associated with congenital heart disease

(1) Eisenmenger syndrome Includes all large intra- and extracardiac defects that begin as systemic-to-pulmonary shunts and progress to severely elevated PVR and to reverse (pulmonary-to-systemic) or bidirectional shunting. Cyanosis, secondary erythrocytosis, and multiple organ involvement are usually present. Closing the defects is contraindicated.
(2) PAH associated with prevalent systemic-to-pulmonary shunts • Correctable[a] • Non-correctable Include moderate-to-large defects. PVR is mildly to moderately increased and systemic-to-pulmonary shunting is still prevalent, whereas cyanosis at rest is not a feature.
(3) PAH with small/coincidental[b] defects Markedly elevated PVR in the presence of cardiac defects considered haemodynamically non-significant (usually ventricular septal defects <1 cm and atrial septal defects <2 cm of effective diameter assessed by echocardiography), which themselves do not account for the development of elevated PVR. The clinical picture is very similar to IPAH. Closing the defects is contraindicated.
(4) PAH after defect correction Congenital heart disease is repaired, but PAH either persists immediately after correction or recurs/develops months or years after correction in the absence of significant, post-operative, haemodynamic lesions.

IPAH, idiopathic pulmonary arterial hypertension; PAH, pulmonary arterial hypertension; PVR, pulmonary vascular resistance.

[a]With surgery or intravascular percutaneous procedure, see also the *Recommendation Table 18* for shunt closure.

[b]The size applies to adult patients. However, also in adults, the simple diameter may be insufficient for defining the haemodynamic relevance of the defect, and also the pressure gradient, the shunt size and direction, and the pulmonary-to-systemic flows ratio should be considered.

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7.5.1. Diagnosis and risk assessment

The diagnostic work-up of PAH associated with adult CHD should be based on the presence of symptoms and includes medical history, physical examination, PFTs, ABG, imaging (especially echocardiography), and exercise and laboratory testing. Of note, standard echocardiographic criteria for detecting PH may not be applicable in complex adult CHD.[559] Right heart catheterization with compartmental oximetry for calculating pulmonary blood flow/systemic blood flow (Qp/Qs) is required to confirm PAH diagnosis and guide therapeutic interventions. Thermodilution should be avoided in the presence of intracardiac shunts, and direct Fick is the most accurate method. Pulmonary vascular resistance may be overestimated due to erythrocytosis.[560] Interpreting invasive haemodynamics (see *Section 5.1.12*) should be made in the context of multiparametric assessment of exercise capacity, laboratory testing, and imaging.

Predictors of worse outcomes in adult CHD-associated PAH are WHO-FC III–IV, exercise intolerance assessed by 6MWD or peak VO_2, history of hospitalization for right HF, biomarkers (NT-proBNP >500 pg/mL, C-reactive protein >10 mg/mL, high serum creatinine, and low albumin levels), iron deficiency, and echocardiographic indices of RV dysfunction.[559,561] When compared with patients with IPAH, patients with Eisenmenger syndrome may have a relatively stable long-term clinical course. The right ventricle is unloaded by the right-to-left shunt, sustaining CO at the expense of hypoxaemia and cyanosis. However, due to immortal time bias, prognosis of Eisenmenger syndrome is not as favourable as previously thought.[562]

As in other forms of PAH, risk assessment is important to guide therapy, and specific risk factors have been described in Eisenmenger syndrome. A large multicentre study showed that mortality in adults with Eisenmenger syndrome was predicted by the presence of pretricuspid shunt, advancing age, low rest oxygen saturation, absence of sinus rhythm, and presence of pericardial effusion.[563]

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7.5.2. Therapy

Outcomes in adult CHD-associated PAH have improved with the availability of new PAH therapies, advances in surgical and perioperative management, and a team-based, multidisciplinary approach in PH centres. These patients should be managed by specialized health professionals. Patient education, behavioural modifications, and social and psychological support are all important aspects of management.

Shunt closure (surgical or interventional) may only be considered in patients with prevalent systemic-to-pulmonary shunting without significantly increased PVR. Criteria for defect closure based on Qp/Qs ratio and (baseline and/or after targeted PAH treatment) PVR have been proposed by the 2020 ESC Guidelines for the management of adult congenital heart disease.[101] Decisions on shunt closure should not be made on haemodynamic numbers alone, and a multiparametric strategy should be followed. For instance, shunt closure is not indicated in the case of desaturation during exercise in the 6MWT or CPET, or when there is secondary erythrocytosis suggesting dynamic reversal of shunt. There is no evidence for a long-term benefit of a treat-and-repair approach in patients with adult CHD-associated PAH with prevalent systemic-to-pulmonary shunts; therefore, there is a need for future prospective studies.[564] Defect closure is contraindicated in all patients with Eisenmenger syndrome, and may also adversely affect patients with small/coincidental defects that behave similarly to IPAH.[565] There are no prospective data available on the usefulness of vasoreactivity testing, balloon closure testing, or lung biopsy for assessing operability and normalization of PVR after closure.[566]

Patients with adult CHD-associated PAH may present with clinical deterioration in different circumstances, such as arrhythmia, during non-cardiac surgery requiring general anaesthesia, dehydration or bleeding, thrombo-embolism, and lung infections. Surgeries should be limited to those deemed essential, and performed in specialized centres with anaesthetists experienced in adult CHD and PAH. Endocarditis should be suspected in patients with sepsis, whereas a cerebral abscess should be excluded in those with neurological symptoms or new headache, especially in those with low oxygen saturations and complex anatomies. It is recommended to avoid strenuous exercise, but mild and moderate activities seem to be beneficial.[567] Patients should receive all recommended vaccinations and endocarditis prophylaxis in the presence of cyanosis. Although pregnant patients with left-to-right shunts and stable, well-controlled PAH have tolerated pregnancy well under specialized care, pregnancy is still associated with both high maternal mortality and foetal complications in Eisenmenger syndrome and should be discouraged in this setting;[568,569] hence, effective contraception is highly recommended. Levonorgestrel-based, long-acting, reversible contraception implants or intrauterine devices have been recommended for these patients.[570]

Secondary erythrocytosis is beneficial for adequate oxygen transport and delivery, and routine phlebotomy should be avoided whenever possible. Symptoms of hyperviscosity in the presence of haematocrit >65% should be approached with appropriate hydration. Iron deficiency should be corrected. When i.v. iron supplementation is administered, special care should be taken to avoid air emboli during administration.[571] Supplemental oxygen therapy has not been shown to impact survival.

Oral anticoagulant treatment with VKAs may be considered in patients with large PA aneurysms with thrombus, atrial arrhythmias, and previous thrombo-embolic events, but with low bleeding risk. In patients with very high Hb levels (>20 mg/dL), standard international normalized ratio measures are less accurate, and citrate-adjusted blood bottles must be used. Regarding using novel oral anticoagulants (NOACs), a large, nationwide, German, adult CHD database (including 106 NOAC-treated patients with Eisenmenger syndrome) showed that NOAC users had higher long-term risk of bleeding, major adverse cardiovascular events, and mortality compared with those on VKAs, suggesting that initiating NOACs should be reserved for experienced adult CHD centres, carefully weighing potential benefits and risks.[572,573]

Compared with other group 1 subgroups, limited data exist on the use of drugs approved for PAH in patients with adult CHD-associated PAH. Bosentan improved 6MWD and decreased PVR in patients with Eisenmenger syndrome in WHO-FC III.[574] Patients with more complex lesions were less likely to respond to PAH therapies compared with patients with simple lesions. An RCT investigating the efficacy of macitentan found no effect on 6MWD in a mixed cohort of patients with Eisenmenger syndrome (6MWD improved in both treatment and placebo arms), although decreases in NT-proBNP and PVR were noted in the macitentan arm.[575]

Experiences with other ERAs and PDE5is have shown favourable functional and haemodynamic results in Eisenmenger syndrome.[576] In a small, single-centre, pilot study, adding nebulized iloprost to a background of oral PAH therapy failed to improve 6MWD in Eisenmenger syndrome.[577] In case symptoms persist or in clinical deterioration, a sequential and symptom-orientated treatment strategy is recommended in Eisenmenger syndrome, starting with an oral ERA (or PDE5i) and escalating therapy. Should symptoms not adequately improve with oral therapies, i.v./s.c. options should be proactively considered.[578] There is a theoretical risk of paradoxical embolism in right-to-left shunt lesions with the presence of a central venous catheter for i.v. therapy; therefore, s.c. prostacyclin analogue infusion may be considered.

The effect of PAH therapies in patients with prevalent systemic-to-pulmonary shunts is less well established. Patients with small/coincidental defects should be treated with PAH medication.[557] This is also the case for patients with PAH after defect correction who have increased mortality compared with those with Eisenmenger syndrome.[579] These patients were included in major RCTs with PAH therapies and should be evaluated based on comprehensive risk assessment (*Table 16*).[580] The effect of PAH therapies in patients with segmental PH remains a matter of debate.[101,581] While some series have reported promising results, there have been cases where therapies were not tolerated.[581] Similarly, using PAH therapies in Fontan circulation has yielded conflicting results, and results of further studies are awaited.[582–584]

Heart–lung transplantation or LTx with heart surgery is an option in highly selected cases not responsive to medical treatment; however, it is limited by organ availability and lesion complexity. Mortality is high during the first year after surgery, especially after heart–lung transplantation, but remains relatively low thereafter.[585]

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Recommendation Table 18 — Recommendations for shunt closure in patients with pulmonary–systemic flow ratio >1.5:1 based on calculated pulmonary vascular resistance

Recommendations	Class[a]	Level[b]
In patients with an ASD, VSD, or PDA and a PVR <3 WU, shunt closure is recommended	I	C
In patients with an ASD, VSD, or PDA and a PVR of 3–5 WU, shunt closure should be considered	IIa	C
In patients with an ASD and a PVR >5 WU that declines to <5 WU with PAH treatment, shunt closure may be considered	IIb	C
In patients with a VSD or PDA and a PVR >5 WU, shunt closure may be considered after careful evaluation in specialized centres	IIb	C
In patients with an ASD and a PVR >5 WU despite PAH treatment, shunt closure is not recommended	III	C

ASD, atrial septal defect; PAH, pulmonary arterial hypertension; PDA, patent ductus arteriosus; PVR, pulmonary vascular resistance; VSD, ventricular septal defect; WU, Wood units.
Decisions on shunt closure should not be made on haemodynamic numbers alone; a multiparametric strategy should be followed (see *Section* 7.5.2).
[a]Class of recommendation.
[b]Level of evidence.

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Recommendation Table 19 — Recommendations for pulmonary arterial hypertension associated with adult congenital heart disease

Recommendations	Class[a]	Level[b]
Risk assessment		
Risk assessment is recommended for patients with persistent PAH after defect closure	I	C
Risk assessment should be considered in patients with Eisenmenger syndrome	IIa	C
Treatment		
Bosentan is recommended in symptomatic patients with Eisenmenger syndrome to improve exercise capacity[574]	I	B
In patients with Eisenmenger syndrome, the use of supplemental oxygen therapy should be considered in cases where it consistently increases arterial oxygen saturation and reduces symptoms	IIa	C
Supplemental iron treatment should be considered in patients with iron deficiency	IIa	C
In patients with adult CHD, including Eisenmenger syndrome, other ERAs, PDE5is, riociguat, prostacyclin analogues, and prostacyclin receptor agonists should be considered	IIa	C
In patients with PAH after corrected adult CHD, initial oral combination therapy with drugs approved for PAH should be considered for patients at low and intermediate risk, while initial combination therapy including i.v./s.c. prostacyclin analogues should be considered for patients at high risk	IIa	C[c]
In patients with adult CHD, including Eisenmenger syndrome, sequential combination therapy should be considered if patients do not meet treatment goals	IIa	C
In the absence of significant haemoptysis, oral anticoagulant treatment may be considered in patients with Eisenmenger syndrome with pulmonary artery thrombosis	IIb	C
In women with Eisenmenger syndrome, pregnancy is not recommended	III	C
In patients with Eisenmenger syndrome, routine phlebotomy to lower elevated haematocrit is not recommended	III	C

CHD, congenital heart disease; ERA, endothelin receptor antagonist; i.v., intravenous; PAH, pulmonary arterial hypertension; PDE5i, phosphodiesterase 5 inhibitor; s.c., subcutaneous.
[a]Class of recommendation.
[b]Level of evidence.
[c]Level of evidence differs from the 2020 ESC Guidelines for the management of adult congenital heart disease because the number of patients with adult CHD included in the AMBITION study was very low.

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7.6. Pulmonary arterial hypertension associated with schistosomiasis

Schistosomiasis is one of the most common chronic infectious diseases worldwide, affecting around 200 million people.[586,587] Schistosomiasis-associated PAH is present in 5% of patients with the hepatosplenic form of the disease.[586] It is thus a leading cause of PAH, especially in some regions of South America, Africa, and Asia. Compared with patients with IPAH, patients with schistosomiasis-associated PAH present with higher CO and lower PVR, and have a better survival.[587] Registry data suggest that survival in schistosomiasis-associated PAH has improved in recent years with the use of PAH drugs.[588]

7.7. Pulmonary arterial hypertension with signs of venous/capillary involvement

The common risk factors, identical genetic substrate, and indistinguishable clinical presentations of PCH and PVOD necessitate their consideration as a single disease belonging to the group 1 PH spectrum of diseases (PAH with signs of venous/capillary involvement).[1,425,589] In PVOD/PCH, post-capillary lesions affecting septal veins and pre-septal venules consist of loose, fibrous remodelling of the intima that may totally occlude the lumen.[1,425,589,590] These changes are frequently associated with PCH consisting of capillary

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ectasia and proliferation, with doubling and tripling of the alveolar septal capillary layers that may be focally distributed within the alveolar interstitium.[425,590]

The proportion of patients with IPAH that fulfil the criteria for PVOD/PCH is ~10%, resulting in a lowest estimate of PVOD/PCH incidence and prevalence of <1 case/million.[425] In contrast to IPAH, there is a male predominance in PVOD/PCH and its prognosis is worse.[425,589,591] Familial PVOD/PCH typically occurs in the young siblings of one generation, with unaffected and sometimes consanguineous parents, indicating that the disease segregates as a recessive trait.[158,425,591] Biallelic mutations in the *EIF2AK4* gene cause heritable PVOD/PCH.[158] In addition, PVOD/PCH can complicate the course of associated conditions, such as SSc,[425] or be associated with exposure to environmental triggers, such as alkylating agents (cyclophosphamide, mitomycin C)[34] and solvents (trichloroethylene).[38]

7.7.1. Diagnosis

Most patients complain of non-specific dyspnoea on exertion and fatigue.[590] Physical examination may reveal digital clubbing and bibasal crackles on lung auscultation.[590] Pulmonary arterial hypertension and PVOD/PCH share the same haemodynamic profile as precapillary PH.[590,591] The PAWP is not elevated because the pulmonary vascular changes occur in small venulae and capillaries, while the LA filling pressure remains normal.[590,591] A diagnosis of PVOD/PCH is based on the results of tests suggesting venous post-capillary involvement, chronic interstitial pulmonary oedema, and capillary proliferation.[1,590,591] These tests include PFTs (decreased DLCO, frequently <50% theoretical values), ABG (hypoxaemia), and non-contrast chest CT (subpleural thickened septal lines, centrilobular ground-glass opacities, and mediastinal lymphadenopathy).[1,425,589,591,592] Importantly, these patients are at risk of drug-induced pulmonary oedema with PAH therapy, a finding suggestive of PVOD/PCH.[425,591] Detecting biallelic *EIF2AK4* mutations is sufficient to confirm a diagnosis of heritable PVOD/PCH.[158,591,592] Lung biopsy is hazardous in PH and is not recommended for diagnosing PVOD/PCH.[1,425]

7.7.2. Therapy

There is no established medical therapy for PVOD/PCH.[425] Compared with IPAH, PVOD/PCH has a poor prognosis and limited response to PAH therapy, with a risk of pulmonary oedema due to pulmonary venous obstruction.[425,591] However, there are reports of incomplete and transient clinical improvement in individual patients with PVOD/PCH treated with PAH therapy, which should be used with great caution in this setting.[425,591] Diuretics, oxygen therapy, and slow titration of PAH therapy can be used on an individual basis.[425] Therefore, therapy for PVOD/PCH should be undertaken at centres with extensive experience in managing PH, and patients should be fully informed about the risks.[425] Anecdotal reports suggest a potential benefit of immunomodulatory treatments, but this approach requires further study.[593] The only curative therapy for PVOD/PCH is LTx, and eligible patients should be referred to a transplant centre for evaluation upon diagnosis.[425,591] Pathological examination of the explanted lungs will confirm the diagnosis.[590]

Recommendation Table 20 — Recommendations for pulmonary arterial hypertension with signs of venous/capillary involvement

Recommendations	Class[a]	Level[b]
A combination of clinical and radiological findings, ABG, PFTs, and genetic testing is recommended to diagnose PAH with signs of venous and/or capillary involvement (PVOD/PCH)[591]	I	A
Identification of biallelic *EIF2AK4* mutations is recommended to confirm a diagnosis of heritable PVOD/PCH[158,591]	I	A
Referral of eligible patients with PVOD/PCH to a transplant centre for evaluation is recommended as soon as the diagnosis is established	I	C
In patients with PVOD/PCH, the use of drugs approved for PAH may be considered with careful monitoring of clinical symptoms and gas exchange	IIb	C
Lung biopsy is not recommended to confirm a diagnosis of PVOD/PCH	III	C

ABG, arterial blood gas analysis; PAH, pulmonary arterial hypertension; PCH, pulmonary capillary haemangiomatosis; PFT, pulmonary function test; PVOD, pulmonary veno-occlusive disease.
[a]Class of recommendation.
[b]Level of evidence.

7.8. Paediatric pulmonary hypertension

Pulmonary hypertension may present at all ages, including in infants and children. Pulmonary hypertension in childhood shares many common features with PH in adulthood; however, there are also important differences, which concern epidemiology, aetiology, genetic background, age-dependent diagnostic and treatment approaches, and disease monitoring. An important and conceptually distinctive feature of paediatric PH is injury to developing foetal, neonatal, or paediatric lung circulation.

7.8.1. Epidemiology and classification

The reported annual incident rate for paediatric PH is 64/million children.[594] The distribution of the various aetiologies of PH in childhood differs from PH in adulthood.[594–596] Pulmonary arterial hypertension is the most frequent type of PH in children, with the vast majority (82%) of cases being infants with transient PAH (i.e. PPHN or repairable cardiac shunt defects). Of the remaining children with PAH, most have either IPAH, HPAH, or irreversible CHD-associated PAH. The reported incidences of IPAH/HPAH and (non-transient) CHD-associated PAH are 0.7 and 2.2/million children, respectively, with a prevalence of 4.4 and 15.6/million children, respectively.[594] Other conditions associated with PAH (*Table 6*) do occur in children but are rare.

Another significant proportion (34–49%) of children with non-transient PH are neonates and infants with PH associated with respiratory disease, especially developmental lung diseases, including bronchopulmonary dysplasia (BPD), congenital diaphragmatic hernia (CDH), and congenital pulmonary vascular abnormalities.[594–598] These children form a prominent and distinctive group in paediatric PH and are currently classified as PH group 3 associated with

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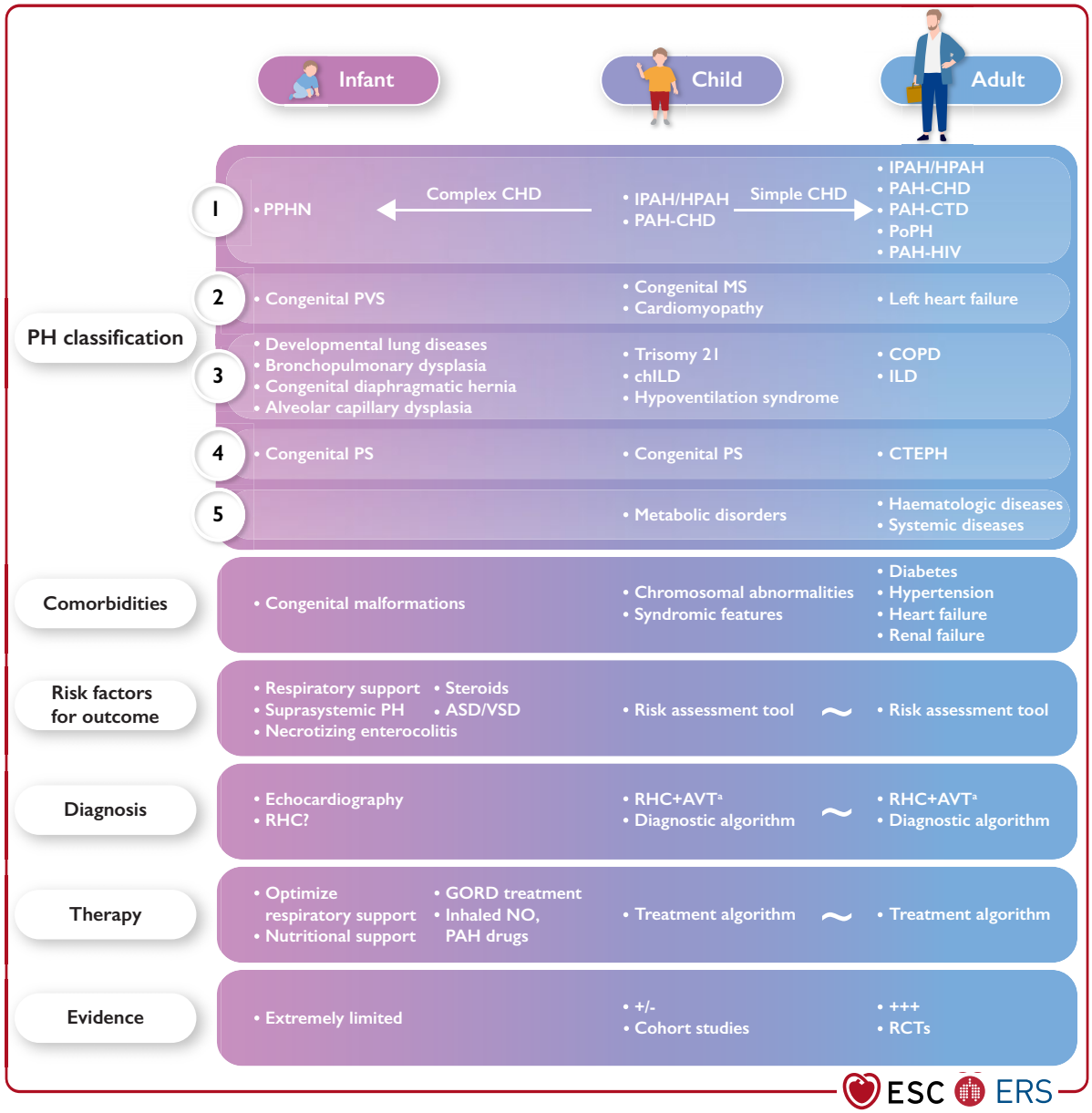


Figure 10 Neonatal and paediatric vs. adult pulmonary hypertension. ASD, atrial septal defect; AVT, acute vasoreactivity testing; CHD, congenital heart disease; COPD, chronic obstructive pulmonary disease; CTEPH, chronic thrombo-embolic pulmonary hypertension; GORD, gastro-oesophageal reflux disease; HPAH, heritable pulmonary arterial hypertension; ILD, interstitial lung disease; IPAH, idiopathic pulmonary arterial hypertension; MS, mitral stenosis; NO, nitric oxide; PAH, pulmonary arterial hypertension; PAH-CHD, PAH associated with congenital heart disease; PAH-CTD, PAH associated with connective tissue disease; PAH-HIV, PAH associated with HIV infection; PH, pulmonary hypertension; PoPH, porto-pulmonary hypertension; PPHN, persistent pulmonary hypertension of the newborn; PS, pulmonary arterial stenosis; PVS, pulmonary vein stenosis; RCT, randomized controlled trial; RHC, right heart catheterization; VSD, ventricular septal defect. [a]In patients with idiopathic, heritable or drug-associated PAH. Pulmonary hypertension in neonates and infants significantly differs in aetiology, pathophysiology, risk assessment, and treatment from older children and adults, while PH in older children has more similarities with PH in adults.

developmental lung disease (*Table 6*; *Table S7*). A significant and growing proportion of children with PH associated with respiratory disease is made up of pre-term infants with BPD. Also, newly recognized genetic developmental lung disorders—including alveolar capillary dysplasia, *TBX4*-mutation-related lung disorders, and surfactant abnormalities—are currently classified in this category (*Figure 10*).[599]

Another distinctive feature of PH in children is the high burden of genetic disorders. Childhood PH is often associated with chromosomal, genetic, and syndromic anomalies (11–52%). Like in adults, gene mutations implicated in the pathogenesis of HPAH are found in 20–30% of sporadic cases, where paediatric HPAH seems to be characterized by an enrichment in *TBX4* and *ACVRL1*

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variations.[600,601] Additionally, 17% of children with PAH have other disorders known to be associated with PAH, including trisomy 21. Finally, 23% of children with PAH have copy number variations not previously associated with PH.[600,602,603]

Given the frequent association of paediatric PAH with chromosomal, genetic, and syndromic anomalies (for which the mechanistic basis for PAH is generally uncertain), genetic testing may be considered for defining aetiology and comorbidities, stratifying risk, and identifying family members at risk; however, this should be after appropriate expert genetic counselling for the child and family (see *Section 5.1.13*).

The clinical PH classification (*Table 6*) is also followed for paediatric PH. To improve applicability of this classification in infants and children with PH, it has been adapted to give room to PH associated with various congenital cardiovascular and pulmonary diseases or specific paediatric conditions (*Tables S5–S8*).[599]

7.8.2. Diagnosis and risk assessment

Historically, the definition of PH in children aged >3 months has been the same as in adults. The definition for PH has now been redefined to mPAP >20 mmHg in adults as well as in children. The impact of an mPAP 21–24 mmHg on outcomes in children is unknown. However, in the interest of consistency and to facilitate transition from paediatric to adult PH care, it is recommended that the updated definition for PH also be followed in children. No treatment recommendations currently exist for this group of children (mPAP 21–24 mmHg).

Regarding the newly introduced criterion to include PVR >2 WU to identify pre-capillary PH in adults, PVR had previously been included in the definition for PAH in children. In children, blood flows are traditionally indexed assuming that systemic and pulmonary blood flows change proportionally with body size, while the transpulmonary pressure gradient does not. Since blood flow is the denominator in the equation for calculating PVR, the need for indexing of PVR in children is emphasized, and the criterion of pulmonary vascular resistance index (PVRI) $\geq$3 WU·m^2 in the definition for PAH in children remains unchanged.[599]

Since the aetiology of paediatric PH is very diverse, a methodical and comprehensive diagnostic approach is crucial to reach an accurate diagnosis and treatment plan. As in adults, IPAH is a diagnosis 'per exclusion'. A diagnostic work-up, similar to that in adults but customized for paediatric PH, is recommended.[599] Pre-term infants with BPD should be screened for PH, since PH is prevalent in this population and seriously affects outcome.[604]

Also in children, RHC is the gold standard for definitively diagnosing and establishing the nature of PH, and provides important data for stratifying risk.[604a,605] To identify those suitable for high-dose CCB treatment, acute vasoreactivity testing is recommended in children with IPAH/HPAH. The criteria used in adults for a positive acute response have identified children who will show sustained benefit from CCB therapy; however, these criteria do not define reversibility of PAH or operability in children with CHD. Since RHC in children with PH may be associated with major complications (in 1–3% of cases, especially in young infants and those in worse clinical condition), risks and benefits have to be balanced in the individual child.[605] Heart catheterization in children with PAH should be exclusively performed in experienced paediatric PH centres. Indications for repeated RHC in children with PH are currently not well defined.

Treatment of children with PAH is based on risk stratification.[599] Predictors of worse outcome in paediatric PAH are similar to those in adults, and include clinical evidence of RV failure, progression of symptoms, WHO-FC III–IV, certain echocardiographic parameters (e.g. TAPSE), and elevated serum NT-proBNP. A 6MWD <350 m has also been suggested as a predictor of worse outcome in paediatric PH, but its value in young children is less established. Further prognosticators identified in paediatric PAH are failure to thrive and haemodynamic variables, such as RAP >10 mmHg, the ratio of mean pulmonary-to-systemic blood pressure >0.75, and PVRI >20 WU·m^2.[602,606,607] Paediatric risk-assessment tools based on these parameters have been retrospectively validated in observational paediatric registries.[599,604a]

7.8.3. Therapy

The ultimate goal of treatment should be to improve survival and facilitate normal childhood activities without limitations. In the absence of RCTs in paediatric PAH, recommended treatment algorithms are extrapolated from those in adults and enhanced with data from observational studies in children with PAH.[599]

Observational cohort studies support treatment algorithms designed for adults to be used for children (including the superiority of combination therapy over monotherapy).[608] Drugs investigated in children, with or without formal approval by the European Medicines Agency (EMA) for treating children with PAH, are shown in *Table 22*.

A paediatric treatment algorithm, derived from that for adults, is based on risk stratification, recommending general measures, high-dose CCB therapy for responders to acute vasoreactivity testing (where close follow-up is mandatory, as some patients may fail long-term therapy), oral or inhaled combination therapy for children at low risk, and combination therapy with i.v./s.c. prostacyclin analogues for those at high risk.[599]

In the case of insufficient response to recommended drug therapy, or when drugs are unavailable, a Potts shunt (a surgical or interventional connection between the left PA and the descending aorta), BAS, or LTx may be considered in children with severe PH (see *Sections 6.3.6.1* and *6.3.8*).[599] Reported clinical experience with Potts shunts is limited to just over 100 patients, predominantly children, with a mortality of 12–25% and long-term clinical benefit in a subset of children with long-term follow-up.[456–459]

Monitoring of treatment effect and disease course is pivotal in managing all patients with PAH (adults and in children). In children with PAH, clinical risk scores including WHO-FC, TAPSE, and serum NT-proBNP are potential treatment targets for goal-orientated treatment.[604a,609]

Contemporary treatment algorithms for infants with PPHN have been proposed but are outside the scope of these guidelines.[610]

The recommendations discussed above apply to children with PAH, whereas the specific group of infants with neonatal PVD, mostly classified as PH associated with developmental lung disease and with heterogeneous aetiology, require a distinct and customized approach (*Figure 10*).

In pre-term infants with BPD and PH, the underlying lung disease should primarily be treated. Frequently, these infants are additionally

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Table 22 Use of pulmonary arterial hypertension therapies in children

Drug	Paediatric study data	European Medicines Agency approval for use in children with PAH	Ref.
Phosphodiesterase 5 inhibitors (oral)			
Sildenafil	RCT, open-label extension: tolerability, efficacy	Yes, for ≥1 year of age Recommended dosing: <20 kg: 30 mg/day in 3 doses; ≥20 kg: 60 mg/day in 3 doses Avoid higher dosing in children (>3 mg/kg/day)	613,614
Tadalafil	RCT, open-label: safety, tolerability, pharmacokinetics	No Suggested dosing: 0.5–1 mg/kg/day in one dose Max: 40 mg/day Evaluated only in children aged >3 years	615,616
Endothelin receptor antagonists (oral)			
Bosentan	Open-label, uncontrolled: safety, tolerability, pharmacokinetics, efficacy	Yes, for ≥1 year of age Paediatric formulation Recommended dosing: 4 mg/kg/day in 2 doses Max: 250 mg/day	617–620
Ambrisentan	Open-label, uncontrolled: safety, tolerability, pharmacokinetics	Yes, for children aged >8 years Recommended dosing: 2.5–10 mg/day in one dose	621,622
Macitentan	Insufficient data in children Open-label, ongoing: efficacy, safety, pharmacokinetics in children aged 2–18 years	No	
Prostacyclin analogues (i.v./s.c.)			
Epoprostenol i.v.	Cohort studies, retrospective	No Suggested dosing: Starting dose: 1–2 ng/kg/min without a known maximum In children, a stable dose is usually 40–80 ng/kg/min Dose increases may be required	623–626
Treprostinil i.v./s.c.	Cohort studies, retrospective: pharmacokinetics	No Suggested dosing: Starting dose: 2 ng/kg/min without a known maximum In children, a stable dose is usually 50–100 ng/kg/min Dose increases may be required	624,626,627
Other			
Iloprost (inhaled)	Insufficient data in children Small case series, retrospective	No	
Selexipag (oral)	Insufficient data in children Randomized, placebo-controlled, add-on, ongoing: safety, tolerability, pharmacokinetics in children aged 2–18 years	No	
Riociguat (oral)	Insufficient data in children Open-label, ongoing: safety, tolerability, pharmacokinetics in children aged 6–18 years	No	

i.v., intravenous; PAH, pulmonary arterial hypertension; RCT, randomized controlled trial; s.c., subcutaneous.

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treated with therapies for PAH, including sildenafil and bosentan; however, these are not approved by the EMA for use in infants with group 3 PH and developmental lung diseases (BPD, CDH). Their effects on outcomes in this population are unclear, and data enabling robust treatment recommendations are lacking. These children should be treated by multidisciplinary teams involving cardiologists, neonatologists, pulmonologists, and nutritionists. Pulmonary hypertension in these infants may disappear with lung healing, although long-term cardiovascular sequelae have been reported.[611,612]

Recommendation Table 21 — Recommendations for paediatric pulmonary hypertension

Recommendations	Class[a]	Level[b]
Children		
It is recommended to perform the diagnostic work-up, including RHC and acute vasoreactivity testing, and treat children with PH at centres with specific expertise in paediatric PH	I	C
In children with PH, a comprehensive work-up for confirming diagnosis and specific aetiology is recommended (similar to that in adults, but adapted for age)	I	C
For confirming PH diagnosis, RHC is recommended, preferably before initiating any PAH therapy	I	C
In children with IPAH/HPAH, acute vasoreactivity testing is recommended to detect those who may benefit from CCB therapy	I	C
It is recommended to similarly define a positive response to acute vasoreactivity testing in children and adults by a reduction in mPAP ≥10 mmHg to reach an absolute value of mPAP ≤40 mmHg, with an increased or unchanged CO	I	C
In children with PAH, a therapeutic strategy based on risk stratification and treatment response is recommended, extrapolated from that in adults but adapted for age	I	C
It is recommended to monitor the treatment response in children with PAH by serially assessing a panel of data derived from clinical assessment, echocardiographic evaluation, biochemical markers, and exercise tolerance tests	I	C
Achieving and maintaining a low-risk profile should be considered as an adequate treatment response for children with PAH	IIa	C
Infants		
It is recommended to screen infants with bronchopulmonary dysplasia for PH[628,629]	I	B
In infants with (or at risk of) bronchopulmonary dysplasia and PH, treating lung disease—including hypoxia, aspiration, and structural airway disease—and optimizing respiratory support is recommended before initiating PAH therapy[630]	I	B
In neonates and infants, a diagnostic and therapeutic approach to PH distinct from that in older children and adults should be considered, given the frequent association with developmental vascular and parenchymal lung disease	IIa	C

Continued

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CCB, calcium channel blocker; CO, cardiac output; HPAH, heritable pulmonary arterial hypertension; IPAH, idiopathic pulmonary arterial hypertension; mPAP, mean pulmonary arterial pressure; PAH, pulmonary arterial hypertension; PH, pulmonary hypertension; RHC, right heart catheterization.
[a]Class of recommendation.
[b]Level of evidence.

8. Pulmonary hypertension associated with left heart disease (group 2)

8.1. Definition, prognosis, and pathophysiology

Among patients with LHD, PH and RV dysfunction are frequently present and associated with high mortality.[47] This includes patients with HF with reduced, mildly reduced, or preserved ejection fraction (HFrEF, HFmrEF, or HFpEF), left-sided valvular heart disease, and congenital/acquired cardiovascular conditions leading to post-capillary PH.[13,631–635] Arguably, PH-LHD represents the most prevalent form of PH, accounting for 65–80% of cases.[47]

Consistent with the general definitions of PH, PH-LHD (group 2 PH) is defined by an mPAP >20 mmHg and a PAWP >15 mmHg. Within this haemodynamic condition of post-capillary PH, IpcPH is defined by PVR ≤2 WU and CpcPH by PVR >2 WU (*Table 5*). The diastolic pressure gradient (DPG) (calculated as the difference between dPAP and PAWP) is no longer used to distinguish between IpcPH and CpcPH because of conflicting data on prognostication in LHD.[142]

Across the spectrum of LHD, increases in PAP and PVR are associated with an increased disease burden and a worse outcome.[13,631,633,635] In a large patient cohort—predominantly with post-capillary PH—a PVR ≥2.2 WU was associated with adverse outcomes and considered abnormal.[13] However, even within this subgroup of patients with LHD and CpcPH, the risk of mortality increases with progressive elevation in PVR. In patients with advanced HFrEF and those with HFpEF or valvular heart disease, a PVR >5 WU carries additional prognostic information and is considered clinically meaningful by physicians.[142,450,631–639] Elevated PVR also appears to be associated with decreased survival in special situations, such as in patients undergoing interventions for correcting valvular heart disease,[634] heart transplantation,[142,633] or LVAD implantation.[142,637] Based on available data, a PVR >5 WU may indicate a severe pre-capillary component, the presence of which may prompt physicians to refer patients to PH centres for specialized care.

The prevalence of PH in patients with LHD is difficult to assess and depends on the methodology of diagnostic testing (echocardiography or invasive haemodynamics), cut-off values used to define PH, and populations studied. Observational studies suggest an estimated prevalence of PH of 40–72% in patients with HFrEF and 36–83% in those with HFpEF.[48,639–643] When PVR is used to define

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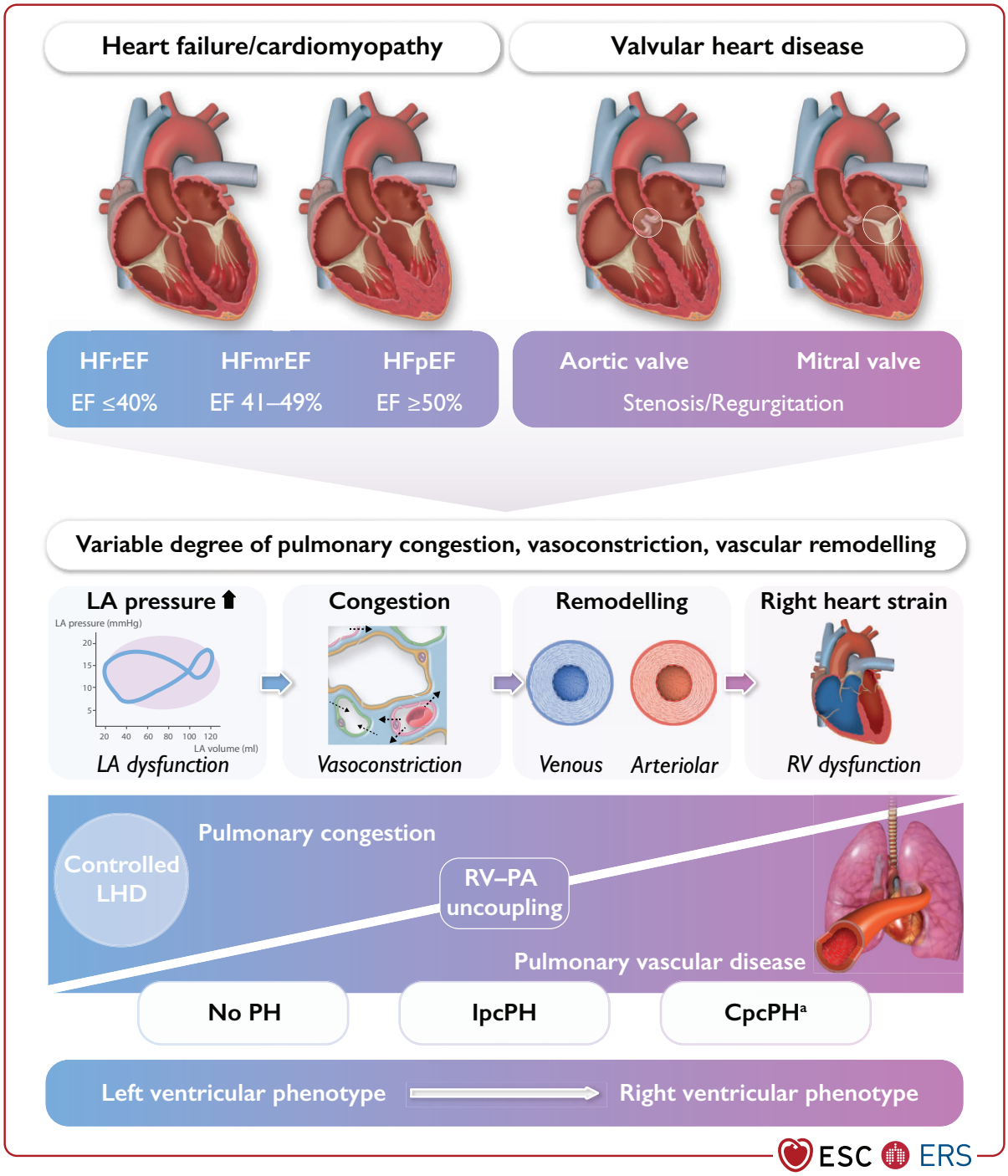


Figure 11 Pathophysiology of pulmonary hypertension associated with left heart disease (group 2). CpcPH, combined post- and pre-capillary pulmonary hypertension; HFmrEF, heart failure with mildly reduced ejection fraction; HFpEF, heart failure with preserved ejection fraction; HFrEF, heart failure with reduced ejection fraction; IpcPH, isolated post-capillary pulmonary hypertension; LA, left atrial; LHD, left heart disease; RV, right ventricle/right ventricular; PA, pulmonary artery; PH, pulmonary hypertension. [a]CpcPH is defined by post-capillary PH and PVR >2 WU; a PVR >5 WU may be considered a severe pre-capillary component.

a pre-capillary component in patients with HF and post-capillary PH, ~20–30% of patients are categorized as having CpcPH.[47,644,645] In patients with valvular heart disease, echocardiographic studies have shown that PH is present in up to 65% of patients with symptomatic aortic stenosis,[646–651] while virtually all patients with severe mitral valve stenosis develop PH,[652] which can also be found in most patients with significant degenerative or functional mitral regurgitation.

The pathophysiology of PH-LHD combines several mechanisms (*Figure 11*): (1) an initial passive increase in LV filling pressures and backward transmission into the pulmonary circulation; (2) PA

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endothelial dysfunction (including vasoconstriction); (3) vascular remodelling (which may occur in both venules and/or arterioles); (4) RV dilatation/dysfunction and functional TR;[653–656] and (5) altered RV–PA coupling.[655–657] The haemodynamic profile of CpcPH vs. IpcPH and elevated PVR reflects pulmonary vascular abnormalities, which contribute to an increased RV afterload. Resulting dysfunction of the RV is frequent and associated with a worse prognosis in patients with PH-LHD. In HFpEF, where RV dysfunction may occur via distinct mechanisms (*Figure S1*), deterioration of RV, but not LV systolic function, has been observed over time, and both prevalent and incident RV dysfunction are predictors of mortality.[658]

The occurrence of PH in patients with LHD may also be due to other causes, including undetected CTEPH or PAH. Further, respiratory comorbidities such as COPD and sleep apnoea are also common in patients with LHD and may contribute to PH and impact prognosis. Patients with HFpEF and PH associated with HFpEF[75,76] may also present with a low DLCO, which is an independent predictor of outcome.[75]

8.2. Diagnosis

In patients with LHD, symptoms (e.g. exertional dyspnoea) and physical signs of PH (e.g. peripheral oedema) frequently overlap with those of the underlying left heart condition and are mostly non-specific. However, while pulmonary congestion or pleural effusion indicate LHD as the underlying cause of PH, other features may suggest the presence of relevant PH (see *Section 5.1.1*).

Routine diagnostic tests including BNP/NT-proBNP, ECG, and echocardiography may show signs of underlying LHD, but may also indicate PH. While BNP/NT-proBNP cannot discriminate between left- or right-sided HF, ECG findings such as right axis deviation or RV strain may suggest the presence of PH in patients with LHD. Echocardiography can diagnose HFrEF and HFpEF; identify specific cardiac conditions, including those with restrictive filling pattern; and diagnose additional valvular heart disease; it may also detect elevated sPAP and other features of PH (RA area, PA enlargement, RV/LV ratio, LV eccentricity index, RV forming the apex), leading to an echocardiographic probability of PH (see *Section 5.1.5*). A stepwise, composite echocardiographic score may discriminate pre- vs. post-capillary PH and predict PVD in patients with LHD.[659,660] Additional information may be gathered from further testing, including biomarkers, imaging-derived markers of RV dysfunction, and CPET-derived variables.[142]

Given the complexity and variability of cardiopulmonary haemodynamics in patients with LHD, the distinction between post- and pre-capillary PH and the diagnosis of PH-LHD vs. other forms of PH can be challenging. Diagnostic clues in the evaluation of suspected PH in LHD include: (1) diagnosis and control of the underlying LHD; (2) evaluation for PH and patient phenotyping; and (3) invasive haemodynamic evaluation, when indicated.

8.2.1. Diagnosis and control of the underlying left heart disease

Patients with suspected PH-LHD will have an established diagnosis of LHD, such as HFrEF/HFmrEF, HFpEF, valvular heart disease, and/or CHD. The distinction between PH associated with HFpEF and other forms of PH (e.g. PAH, CTEPH) may be challenging, particularly given the increased burden of cardiovascular comorbidities in real-world PAH populations.[142,450,661] In this context, validated scores for diagnosing HFpEF (HFA-PEFF, H2FPEF)[16,662,663] may be helpful for detecting it as an underlying condition in PH, and the presence or absence of risk factors for PAH or CTEPH should be determined. Patients with signs of predominant RV strain and/or PH should be further evaluated. Patients should be assessed or reassessed when they are fully recompensated and in a clinically stable condition.

8.2.2. Evaluation of pulmonary hypertension and patient phenotyping

Patients with LHD and suspected PH should be evaluated following the diagnostic strategy for PH (see *Section 5*). This requires identifying clinical features and a multimodal approach using non-invasive diagnostic tests such as echocardiography, ECG, and BNP/NT-proBNP levels. In the presence of mild PH and predominant LHD, no further testing may be necessary. Otherwise, CTEPH and significant lung disease should be ruled out by V/Q scan and PFTs, and additional cardiac imaging including cMRI may be considered in selected cases. For phenotyping, a combination of variables may help to determine the likelihood of LHD, and HFpEF in particular, vs. other causes of PH (*Table 23*). Pulmonary hypertension associated with left heart disease is likely in the presence of known cardiac disease, multiple cardiovascular comorbidities/risk factors, atrial fibrillation at diagnosis, and specific imaging findings (LV hypertrophy, increased LA size, and reduced LA strain). Although exercise echocardiography has been proposed to uncover HFpEF, it is unable to diagnose or classify PH in this context. A combination of clinical findings and phenotyping is required to decide about the need for further invasive assessment.

8.2.3. Invasive assessment of haemodynamics

The decision to perform cardiac catheterization and to invasively assess cardiopulmonary haemodynamics should depend on the presence of an intermediate to high echocardiographic probability of PH, and should be determined by the need to obtain relevant information for prognostication or management. In patients with a high likelihood of LHD as the main cause of PH, or with established underlying LHD and mild PH (*Table 23*), invasive assessment for PH is usually not indicated. Indications for RHC in LHD include: (1) suspected PAH or CTEPH; (2) suspected CpcPH with a severe pre-capillary component, where further information will aid phenotyping and treatment decisions (*Figure S2*); and (3) advanced HF and evaluation for heart transplantation. While several haemodynamic measures (mPAP, PVR, pulmonary arterial compliance [PAC], transpulmonary pressure gradient, and DPG) are associated with outcomes in PH-LHD,[142,632,635] the most robust and consistent data are available for PVR. Invasive assessment should be conducted in experienced centres, when management of the underlying LHD has been optimized and patients are in a clinically stable condition. With respect to respiratory variations of intrathoracic pressures, all pressure readings should be taken at end-expiration.

Additional testing during RHC may be useful for distinguishing between PAH and HFpEF,[18,23,664–669] and to uncover LHD in patients with a high likelihood of PH-LHD and normal resting PAWP;[670–673] both exercise testing and fluid challenge may be considered in special situations (see *Section 5.1.12*). Conditions associated with reduced LV diastolic compliance or valvular heart disease may be associated

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Table 23 Patient phenotyping and likelihood for left heart disease as cause of pulmonary hypertension

Feature	PH-LHD unlikely	Intermediate probability	PH-LHD likely
Age	<60 years	60–70 years	>70 years
Obesity, hypertension, dyslipidaemia, glucose intolerance/ diabetes	No factors	1–2 factors	>2 factors
Presence of known LHD	No	Yes	Yes
Previous cardiac intervention	No	No	Yes
Atrial fibrillation	No	Paroxysmal	Permanent/persistent
Structural LHD	No	No	Present
ECG	Normal or signs of RV strain	Mild LVH	LBBB or LVH
Echocardiography	No LA dilation E/e' <13	No LA dilation Grade <2 mitral flow	LA dilation (LAVI >34 mL/ m^2) LVH Grade >2 mitral flow
CPET	High VE/VCO_2 slope No EOV	Elevated VE/VCO_2 slope EOV	Mildly elevated VE/VCO_2 slope EOV
cMRI	No left heart abnormalities		LVH LA dilation (strain or LA/RA >1)

cMRI, cardiac magnetic resonance imaging; CPET, cardiopulmonary exercise testing; E/e', ratio between early mitral inflow velocity and mitral annular early diastolic velocity; ECG, electrocardiogram; EOV, exercise oscillatory ventilation; LA, left atrial; LAVI, left atrial volume index; LBBB, left bundle branch block; LHD, left heart disease; LVH, left ventricular hypertrophy; PH, pulmonary hypertension; PH-LHD, left heart disease associated with pulmonary hypertension; RA, right atrium; RV, right ventricle; $VE/VECO_2$, ventilatory equivalents for carbon dioxide.
Assigning the likelihood of LHD as a cause of PH. This assessment may help to decide which patients should undergo a full work-up, including invasive haemodynamic assessment (see *Figure 11* and *Figure S2*).

with a rapid increase in PAWP when challenged with increased systemic venous return.[674] While the upper limit of normal remains controversial,[142,143,665,667] a PAWP cut-off of >18 mmHg has been suggested to identify HFpEF as the underlying cause of PH, despite normal PAWP at baseline.[143] While this may help to classify PH, therapeutic consequences of such testing remain to be determined.

As differentiating between severe PH associated with HFpEF and IPAH with cardiac comorbidities is challenging, patients with an unclear diagnosis, particularly those with a predominant pre-capillary component (e.g. PVR >5 WU), should be referred to a PH centre for individualized management.

8.3. Therapy

The primary strategy in managing PH-LHD is optimizing treatment of the underlying cardiac disease. Nevertheless, a pathophysiological sequence ranging from left-sided heart disease via pulmonary circulation to chronic right heart strain (at rest or exercise) is present in many patients.[47] Since deterioration of RV function over time is associated with poor outcomes in HFpEF,[658] preserving RV function should be considered an important treatment goal. Diuretics remain the cornerstone of medical therapy in the presence of fluid retention due to PH-LHD.

There is limited and conflicting evidence for the use of drugs approved for PAH in patients with group 2 PH. Some medications may have variable and potentially detrimental effects in such patients and are therefore not indicated in PH-LHD. Management strategies for PH in various left heart aetiologies are described below.

8.3.1. Pulmonary hypertension associated with left-sided heart failure

8.3.1.1. Heart failure with reduced ejection fraction

Patients with HFrEF or HFmrEF require guideline-directed treatment including established medical and interventional therapies.[27] In patients with advanced HFrEF, implanting an LVAD may significantly reduce or even normalize mPAP,[675] although this is not achieved in all patients,[676] and an increased DPG emerged as a negative prognostic factor after LVAD implantation.[677] With regards to PAH drugs, bosentan was assessed in an RCT of patients with PH associated with HFrEF,[678] showing no efficacy but an increase in adverse events compared with placebo, predominantly related to fluid retention. Small studies have suggested that sildenafil may improve haemodynamics and exercise capacity in PH and HFrEF,[679–681] but RCTs are lacking.

8.3.1.2. Heart failure with preserved ejection fraction

In patients with HFpEF, blood pressure, volume load, and risk factors should be controlled, which may lower filling pressures and PAP.[27] Recently, the SGLT-2i empagliflozin improved outcomes in patients

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with an LV ejection fraction of 40–60%.[682] Endothelin receptor antagonists have not proved successful in this population, as both bosentan[683] and macitentan[684] failed to show efficacy but rather led to more adverse events (fluid retention) vs. placebo in patients with HFpEF-associated PH and HF with ejection fraction >35%-associated CpcPH, respectively. Phosphodiesterase 5 inhibitors were assessed in two small RCTs in patients with HFpEF and PH with distinct haemodynamic characteristics. In patients with a predominantly IpcPH profile, sildenafil had no effect on mPAP (primary endpoint) or other haemodynamic and clinical measures vs. placebo.[685] In patients with a predominantly CpcPH profile, sildenafil improved haemodynamics, RV function, and quality of life at 6 and 12 months vs. placebo.[686] Furthermore, retrospective analyses and registry data suggested improvements in exercise capacity with PDE5i therapy in patients with HFpEF-associated CpcPH and with a severe pre-capillary component (PVR mostly >5 WU).[450,687]

8.3.1.3. Interatrial shunt devices

Recent data suggest that specific interventions may be considered in selected cases of HFpEF, such as interatrial shunt devices to unload the left heart. While this was associated with short-term improvements in pulmonary vascular function,[688] the long-term effect on the pulmonary circulation remains unknown. The recent REDUCE LAP-HF II trial failed to show a reduction in HF events after placement of an atrial shunt device in a population of HF patients with LVEF ≥40%,[689] with worse outcomes in the presence of PVD.[690] In addition, a sustained increase in PA blood flow may be a matter of concern, as this may trigger vascular remodelling in patients with pre-existing PH.

8.3.1.4. Remote pulmonary arterial pressure monitoring in heart failure

The importance of decongestion in patients with HF is underscored by the use of implantable pressure sensors, remotely monitoring PAP as a surrogate of left-sided filling pressure. Pulmonary arterial pressure-based adjustment of HF therapy substantially reduced HF hospitalizations and improved outcomes in both patients with HFpEF and HFrEF,[691–694] with adjustment of diuretic therapy being the most prominent therapeutic consequence. Further strategies to optimize management depending on the haemodynamic phenotype in PH-LHD remain to be established. In HFrEF, novel medical therapies such as ARNIs and SGLT-2is reduced remotely monitored PAP and diuretic use,[695–698] potentially providing opportunities to further optimize PAP-guided HF therapy.

8.3.2. Pulmonary hypertension associated with valvular heart disease

Pulmonary hypertension frequently occurs as a consequence of valvular heart disease. While surgical or interventional approaches for valvular repair improve cardiopulmonary haemodynamics by reducing PAWP and PAP and improving forward SV,[699] persistent PH after correcting valvular heart disease is frequent and associated with adverse outcomes.[634,700]

8.3.2.1. Mitral valve disease

Both mitral stenosis and regurgitation regularly lead to post-capillary PH. Functional (secondary) mitral regurgitation occurs in both HFrEF and HFpEF, and is an important contributor to PH in LHD. Reducing mitral regurgitation according to the recommendations of the 2021 ESC/EACTS Guidelines for the management of valvular heart disease[28] has a crucial role in improving haemodynamics in patients with HFrEF, as this reduces mPAP and PAWP and improves the CI.[699] Nevertheless, registry data have demonstrated that even moderately elevated sPAP negatively impacts post-procedural outcomes after catheter-based therapy.[700]

8.3.2.2. Aortic stenosis

In patients with aortic stenosis undergoing surgical or catheter-based aortic valve repair, pre-interventional PH is associated with a higher risk of in-hospital adverse events and adverse long-term outcomes.[646–651] Although post-procedural improvement in PH correlates with symptom relief and favourable outcomes, persistence of PH is common, and even moderate PH is associated with a higher all-cause mortality.[646–651]

Of note, medical therapy of PH post-valvular repair may be harmful. A randomized study of 231 patients with surgically corrected valvular heart disease and persistent PH showed that sildenafil therapy vs. placebo was associated with worse outcome when compared with placebo;[701] however, this study did not distinguish between different types of PH (pre-capillary, IpcPH, and CpcPH).

8.3.2.3. Tricuspid regurgitation

Severe TR is associated with volume overload, increased RV workload, and maladaptive remodelling, leading to symptomatic right HF and impaired survival.[702,703] While primary TR is relatively rare, functional TR may arise from annular dilation in the presence of both PH and LHD. Transcatheter tricuspid valve interventions have recently emerged, aiming at reducing TR and RV volume overload. Of note, correcting TR in patients with PAH or PH in (non-valvular) LHD with significantly elevated PVR and/or RV dysfunction must be considered with great caution, as this may be hazardous.[704] Right ventricle–PA coupling is an independent predictor of all-cause mortality in such patients.[705] Patient selection appears crucial, and a comprehensive diagnostic approach integrating imaging modalities and invasive haemodynamic assessment is necessary in the evaluation process prior to tricuspid valve repair, particularly since echocardiography underestimates sPAP in the presence of severe TR.

8.3.3. Recommendations on the use of drugs approved for PAH in PH-LHD

The recommendations on the use of drugs approved for PAH in patients with PH-LHD have been established based on key narrative question 5 (Supplementary Data, *Section 8.3*).

The recommendations on the use of PDE5is in patients with CpcPH associated with HFpEF are based on PICO question II (Supplementary Data, *Section 8.4*). Two RCTs that enrolled patients with HFpEF and PH were identified, but no study that specifically enrolled patients with HFpEF and CpcPH. Harmful effects cannot be excluded, even if the available data from clinical studies, case series, and registries suggest that PDE5is may be safely administered to patients with HFpEF-associated CpcPH. As a result, a general recommendation for or against the use of PDE5is in patients with HFpEF and CpcPH cannot be made. However, it is clinically relevant to make a recommendation against their use for patients with HFpEF and IpcPH.

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Recommendation Table 22 — Recommendations for pulmonary hypertension associated with left heart disease

Recommendation Table 22A

Recommendations	Class[a]	Level[b]
In patients with LHD, optimizing treatment of the underlying condition is recommended before considering assessment of suspected PH[27,28]	I	A
RHC is recommended for suspected PH in patients with LHD, if it aids management decisions	I	C
RHC is recommended in patients with severe tricuspid regurgitation with or without LHD prior to surgical or interventional valve repair	I	C
For patients with LHD and suspected PH with features of a severe pre-capillary component and/or markers of RV dysfunction, referral to a PH centre for a complete diagnostic work-up is recommended[29,47,142]	I	C
In patients with LHD and CpcPH with a severe pre-capillary component (e.g. PVR >5 WU), an individualized approach to treatment is recommended	I	C
When patients with PH and multiple risk factors for LHD, who have a normal PAWP at rest but an abnormal response to exercise or fluid challenge, are treated with PAH drugs, close monitoring is recommended	I	C
In patients with PH at RHC, a borderline PAWP (13–15 mmHg) and features of HFpEF, additional testing with exercise or fluid challenge may be considered to uncover post-capillary PH[133,143]	IIb	C
Drugs approved for PAH are not recommended in PH-LHD[c] [631,678,683,684,701,706]	III	A

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Recommendation Table 22B

	GRADE			
Recommendations	Quality of evidence	Strength of recommendation	Class[a]	Level[b]
No recommendation can be given for or against the use of PDE5is in patients with HFpEF and combined post- and pre-capillary PH	Low	None	–	–
The use of PDE5is in patients with HFpEF and isolated post-capillary PH is not recommended	Low	Conditional	III	C

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CpcPH, combined post- and pre-capillary PH; ERA, endothelin receptor antagonist; HF, heart failure; HFpEF, heart failure with preserved ejection fraction; HFrEF, heart failure with reduced ejection fraction; LHD, left heart disease; PAH, pulmonary arterial hypertension; PAWP, pulmonary arterial wedge pressure; PDE5is, phosphodiesterase 5 inhibitors; PH, pulmonary hypertension; PH-LHD, pulmonary hypertension associated with left heart disease; PVR, pulmonary vascular resistance; RHC, right heart catheterization; RV, right ventricular; WU, Wood units.

[a]Class of recommendation.

[b]Level of evidence.

[c]Safety concerns have been identified when ERAs are used in patients with HF (HFpEF and HFrEF, with or without PH) and when sildenafil is used in patients with persistent PH after correction of valvular heart disease.

9. Pulmonary hypertension associated with lung diseases and/or hypoxia (group 3)

Pulmonary hypertension is frequently observed in patients with COPD and/or emphysema, ILD, combined pulmonary fibrosis and emphysema (CPFE), and hypoventilation syndromes.[52,165,707,708] Pulmonary hypertension is uncommon in obstructive sleep apnoea unless other conditions coexist, such as COPD or daytime hypoventilation.[709] At high altitude (>2500 m) hypoxia-induced PH is thought to affect >5% of the population, the development of PH being related to geography and genetic factors.[710]

A PH screening study performed on a large cohort of >100 patients with lymphangioleiomyomatosis confirmed that PH is usually mild in that setting: from six patients (5.7%) presenting with pre-capillary PH, none had mPAP >30 mmHg and PH was associated with PFT alteration, suggesting that the rise in mPAP is associated with parenchymal involvement.[711] Thus, PH in lymphangioleiomyomatosis is now classified in group 3 PH.[1]

In patients with lung disease, PH is categorized as non-severe or severe, depending on haemodynamic findings (*Figure 12*). In the 2015 ESC/ERS Guidelines for the diagnosis and treatment of PH, severe PH was defined by mPAP >35 mmHg or mPAP ≥25 mmHg with CI <2.5 L/min/m^2.[25,26] However, two recent studies have demonstrated that a PVR >5 WU is a better threshold for predicting worse prognosis in patients with PH associated with both COPD and ILD.[712,713] Based on these data, the current guidelines used PVR to distinguish between non-severe PH (PVR

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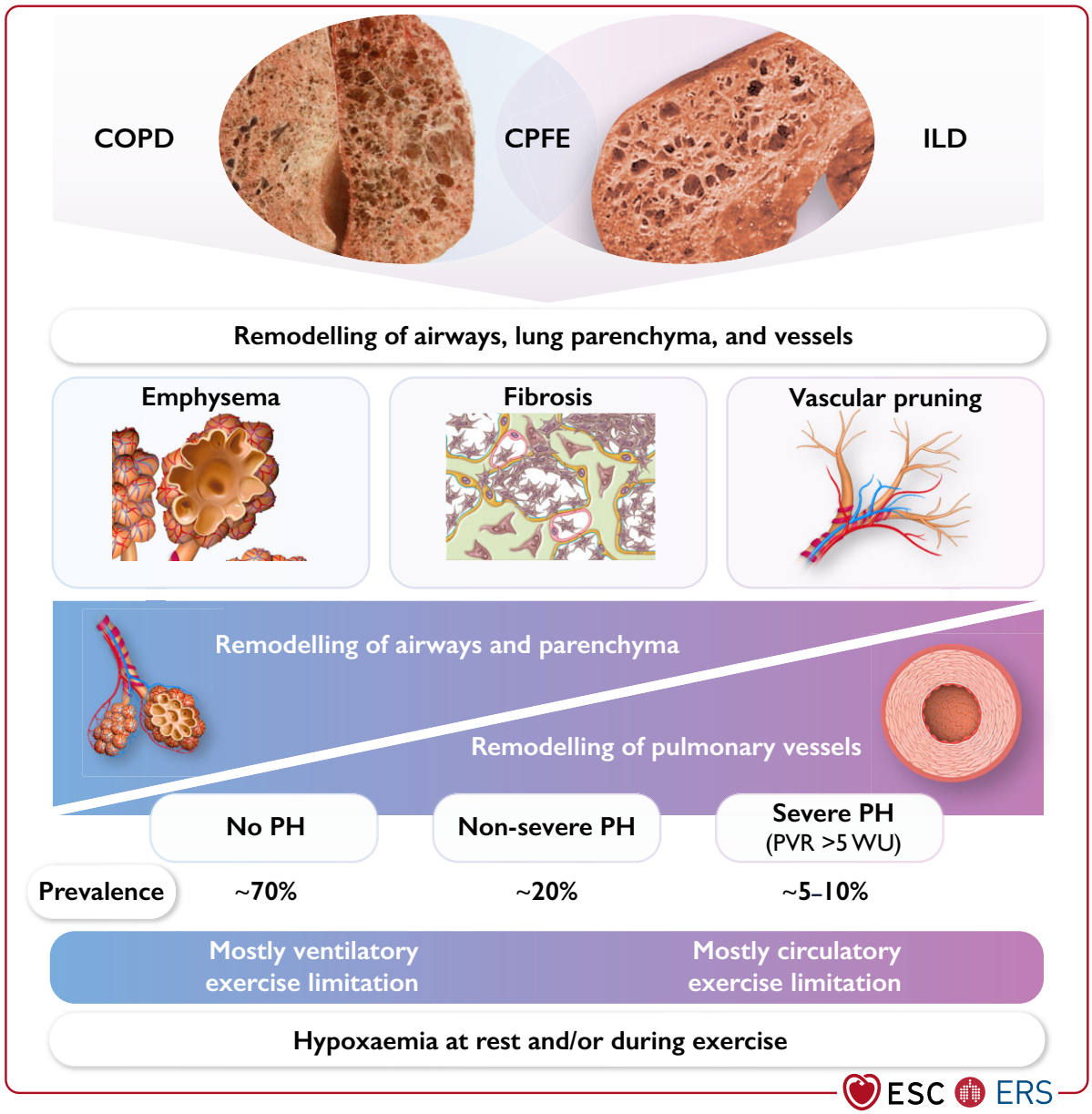


Figure 12 Pathophysiology of pulmonary hypertension associated with lung disease (group 3). COPD, chronic obstructive pulmonary disease; CPFE, combined pulmonary fibrosis and emphysema; ILD, interstitial lung disease; PH, pulmonary hypertension; PVR, pulmonary vascular resistance; WU, Wood units. Shown are underlying lung diseases (**upper panel**); contributing pathogenic pulmonary alterations of airways, parenchyma, and vessels (**middle panel**); and the relation of airway/parenchymal remodelling and vascular remodelling to the degree of PH and its consequences for exercise limitation (ventilatory vs. circulatory, **lower panel**).

≤5 WU) and severe PH (PVR >5 WU). Whereas non-severe PH is common in advanced COPD and ILD defined by spirometric criteria, severe PH is uncommon, occurring in 1–5% of cases of COPD and <10% of patients with advanced ILD, with limited data in obesity hypoventilation syndrome.[714,715] Even non-severe PH in lung disease negatively impacts symptoms and survival, and is associated with increased hospitalization.[715–717] Patients with lung disease and severe PH have a worse outcome than those with non-severe PH, providing evidence that this distinction has clinical significance.[51,712,713,718,719] It is noteworthy that developing severe PH is largely independent of spirometry but usually accompanied by hypoxaemia, low $PaCO_2$, and a significant reduction in DLCO.[51,714,718,719]

Pulmonary hypertension presenting in patients with lung disease may be due to a number of causes, including undiagnosed CTEPH or PAH.[714,720] Cardiac comorbidities are also common in patients with lung disease and may contribute to PH. A number of distinct phenotypes of PH in patients with lung disease, including a pulmonary vascular phenotype, have been proposed.[51,720] The pulmonary vascular phenotype is characterized by better preserved spirometry, low DLCO, hypoxaemia, a range of parenchymal involvement on lung imaging, and a circulatory limitation to exercise.[51,714,718–722] Recent studies have shown that the clinical characteristics, disease trajectory, response to treatment,[451,718,719] and histological correlates[723,724] of patients with severe PH and minor lung disease are different to those in patients with IPAH, including a poorer prognosis.

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9.1. Diagnosis

In patients with lung disease, symptoms of PH, especially exertional dyspnoea, overlap with those of the underlying condition. Physical findings may also be non-specific, for example: ankle swelling is common during episodes of ventilatory failure in COPD, where activation of the renin–angiotensin–aldosterone system may cause fluid retention, usually in the setting of preserved RV function.

Non-invasive tests—such as ECG showing right axis deviation or RV strain, elevated levels of BNP/NT-proBNP, CPET, or features on cross-sectional imaging—may suggest the diagnosis of PH in patients with lung disease.[725,726] Echocardiography remains the most widely used non-invasive diagnostic tool for assessing PH; however, the accuracy of echocardiography in patients with advanced respiratory diseases is low, with a TRV unmeasurable in >50% of patients in some studies, and there is a tendency to overestimate PAP and misclassify patients with PH.[86,87,727] More recent data suggest that a stepwise, composite, echocardiographic score can identify patients with severe PH, with and without an estimate of TRV, using other echocardiographic features including RA area, RV:LV ratio, and LV eccentricity index.[728] Where PH is suspected, combining echocardiography with a contrast-enhanced CT may aid diagnostic assessment and disease classification.[108,729–731] Pulmonary artery enlargement, RV outflow hypertrophy, and increased RV:LV ratio may suggest a diagnosis of PH.[108] Ideally, assessments should be made or repeated when the patient is clinically stable, as exacerbations can significantly raise PAP.

Key parts of evaluating suspected PH in lung disease include integrating: (1) the presence or absence of risk factors for PAH, CTEPH, or LHD; (2) clinical features, including disease trajectory (e.g. rapid recent deterioration vs. gradual change over years, and oxygen requirements); (3) PFTs, including DLCO and blood gas analysis; (4) NT-proBNP measurements, ECG, and echocardiography; and (5) cross-sectional imaging with contrast-enhanced CT, SPECT, or V/Q lung scan and, in selected cases, cMRI[732] to assess the need for RHC. Cardiopulmonary exercise testing may be helpful in assessing ventilatory or cardiac limitation in patients with lung disease,[121,733] although data are limited regarding its clinical use in identifying patients with PH in lung disease.

Indications for RHC in lung disease include assessment for surgical treatments (selected patients considered for LTx and lung volume reduction surgery), suspected PAH or CTEPH, and where further information will aid phenotyping of disease and consideration of therapeutic interventions (*Figure S3*).[712,718,734] Such testing should ideally be conducted in PH centres when patients are clinically stable and treatment of underlying lung disease has been optimized. Consideration should be given to how pressure measurements are made, due to the impact of changing intrathoracic pressures on pulmonary haemodynamics during the respiratory cycle (see *Section 5.1.12*).[735]

9.2. Therapy

The therapeutic approach to group 3 PH starts with optimizing the treatment of the underlying lung disease, including supplementary oxygen and non-invasive ventilation, where indicated, as well as enrolment into pulmonary rehabilitation programmes.[736] There is limited and conflicting evidence for the use of medication approved for PAH in patients with group 3 PH, and these drugs may have variable and sometimes detrimental effects on haemodynamics, exercise capacity, gas exchange, and outcomes in this patient population.[181,737–740]

9.2.1. Pulmonary hypertension associated with chronic obstructive pulmonary disease or emphysema

Studies using drugs approved for PAH in patients with PH associated with COPD or emphysema have yielded conflicting results and are mostly limited by small sample size, short duration, and insufficient haemodynamic characterization of PH.[739,741,742] In a 16 week RCT of 28 patients with COPD and severe PH confirmed by RHC, sildenafil therapy resulted in statistically significant improvements in PVR and quality of life.[743] Registry data identified that ~30% of patients with COPD and severe PH, predominantly treated with PDE5is, had improved WHO-FC, 6MWD, and PVR vs. baseline, and those with a treatment response had improved transplant-free survival.[51,718] However, in the absence of large randomized trials, the evidence is insufficient to support the general use of medication approved for PAH in patients with COPD and PH. Patients with COPD and suspected or confirmed severe PH should be referred to PH centres for individual decision-making.

9.2.2. Pulmonary hypertension associated with interstitial lung disease

Numerous phase 2 and phase 3 studies have investigated the use of ERAs to treat ILD, all with negative results.[740,744,745] In addition, the PDE5i sildenafil has been investigated in phase 3 trials of patients with ILD, also with negative results.[746,747] Few data from RCTs are available for patients with PH associated with ILD, and many of the studies performed for this indication[748,749] suffered from the same limitations as the aforementioned studies in PH associated with COPD. In addition, there were several adverse safety signals: ambrisentan was associated with an increased risk of clinical worsening in patients with ILD with and without PH,[740,750] while riociguat was associated with an increased risk of clinical worsening events, including potential excess mortality, in patients with PH associated with idiopathic interstitial pneumonia.[181]

In contrast, promising results have been obtained with the use of inhaled treprostinil. A phase 3 RCT (INCREASE) examined inhaled treprostinil at a target dose of 72 µg given four times daily in 326 patients with PH associated with ILD.[734,751] The PH diagnosis was confirmed by RHC within 1 year prior to enrolment. At week 16, the placebo-corrected 6MWD improved by 31 m with inhaled treprostinil. There were also improvements in NT-proBNP and clinical worsening events, the latter driven by a lower proportion of patients whose 6MWD declined by >15% from baseline.

Given the significant impact of even non-severe PH in patients with lung disease, eligible patients should be referred for LTx evaluation. In patients with ILD and PH, inhaled treprostinil may be considered based on the findings from the INCREASE study, but further data are needed, especially on long-term outcomes. The routine use of other medication approved for PAH is not recommended in patients with ILD and non-severe PH. For patients with severe PH and/or severe RV dysfunction, or where there is uncertainty regarding the treatment of PH, referral to a PH centre is recommended for careful evaluation, to facilitate entry into RCTs, and consider PAH therapies on an individual basis (*Figure S3*). Registry data show that some patients with group 3 PH are being treated with PAH medication, predominantly PDE5is,[718,752,753] but it is unclear if and to what extent these patients benefit from this treatment.

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9.2.3. Recommendations on the use of drugs approved for PAH in PH associated with lung disease

The recommendations on the use of drugs approved for PAH in patients with PH associated with COPD and ILD have been established based on key narrative questions 6 and 7 (Supplementary Data, *Sections 9.1* and *9.2*, respectively).

The recommendations on the use of PDE5is in patients with severe PH associated with ILD are based on PICO question III (Supplementary Data, *Section 9.3*). There are no direct data from RCTs on the safety, tolerability, and efficacy of PDE5is in patients with PH associated with ILD. The indirect data included in the guidelines do not enable firm conclusions to be drawn. Given the lack of robust evidence, the Task Force members felt unable to provide a recommendation for or against the use of PDE5is in patients with ILD and severe PH, and recommend that these patients are referred to a PH centre for individualized decision-making.

10. Chronic thrombo-embolic pulmonary hypertension (group 4)

All patients whose symptoms can be attributed to post-thrombo-embolic fibrotic obstructions within the PA are

Recommendation Table 23 — Recommendations for pulmonary hypertension associated with lung disease and/or hypoxia

Recommendation Table 23A

Recommendations	Class[a]	Level[b]
If PH is suspected in patients with lung disease, it is recommended that echocardiography[c] be performed and the results interpreted in conjunction with ABG, PFTs including DLCO, and CT imaging	I	C
In patients with lung disease and suspected PH, it is recommended to optimize treatment of the underlying lung disease and, where indicated, hypoxaemia, sleep-disordered breathing, and/or alveolar hypoventilation	I	C
In patients with lung disease and suspected severe PH, or where there is uncertainty regarding the treatment of PH, referral to a PH centre is recommended[d]	I	C
In patients with lung disease and severe PH, an individualized approach to treatment is recommended	I	C
It is recommended to refer eligible patients with lung disease and PH for LTx evaluation	I	C
In patients with lung disease and suspected PH, RHC is recommended if the results are expected to aid management decisions	I	C
Inhaled treprostinil may be considered in patients with PH associated with ILD[734]	IIb	B
The use of ambrisentan is not recommended in patients with PH associated with IPF[740]	III	B
The use of riociguat is not recommended in patients with PH associated with IIP[181]	III	B
The use of PAH medication is not recommended in patients with lung disease and non-severe PH[e]	III	C

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Recommendation Table 23B

	GRADE			
Recommendations	Quality of evidence	Strength of recommendation	Class[a]	Level[b]
PDE5is may be considered in patients with severe PH associated with ILD (individual decision-making in PH centres)	Very low	Conditional	IIb	C
The use of PDE5is in patients with ILD and non-severe PH is not recommended	Very low	Conditional	III	C

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ABG, arterial blood gas analysis; CT, computed tomography; DLCO, Lung diffusion capacity for carbon monoxide; IIP, idiopathic interstitial pneumonia; IPF, idiopathic pulmonary fibrosis; ILD, interstitial lung disease; LTx, lung transplantation; PAH, pulmonary arterial hypertension; PDE5i, phosphodiesterase 5 inhibitor; PFT, pulmonary function test; PH, pulmonary hypertension; RHC, right heart catheterization.

[a]Class of recommendation.

[b]Level of evidence.

[c]Assessments should ideally be made when the patient is clinically stable, as exacerbations can significantly raise pulmonary artery pressure.

[d]This recommendation does not apply to patients with end-stage lung disease who are not considered candidates for LTx.

[e]This does not include inhaled treprostinil, which may be considered in patients with PH associated with ILD, irrespective of PH severity.

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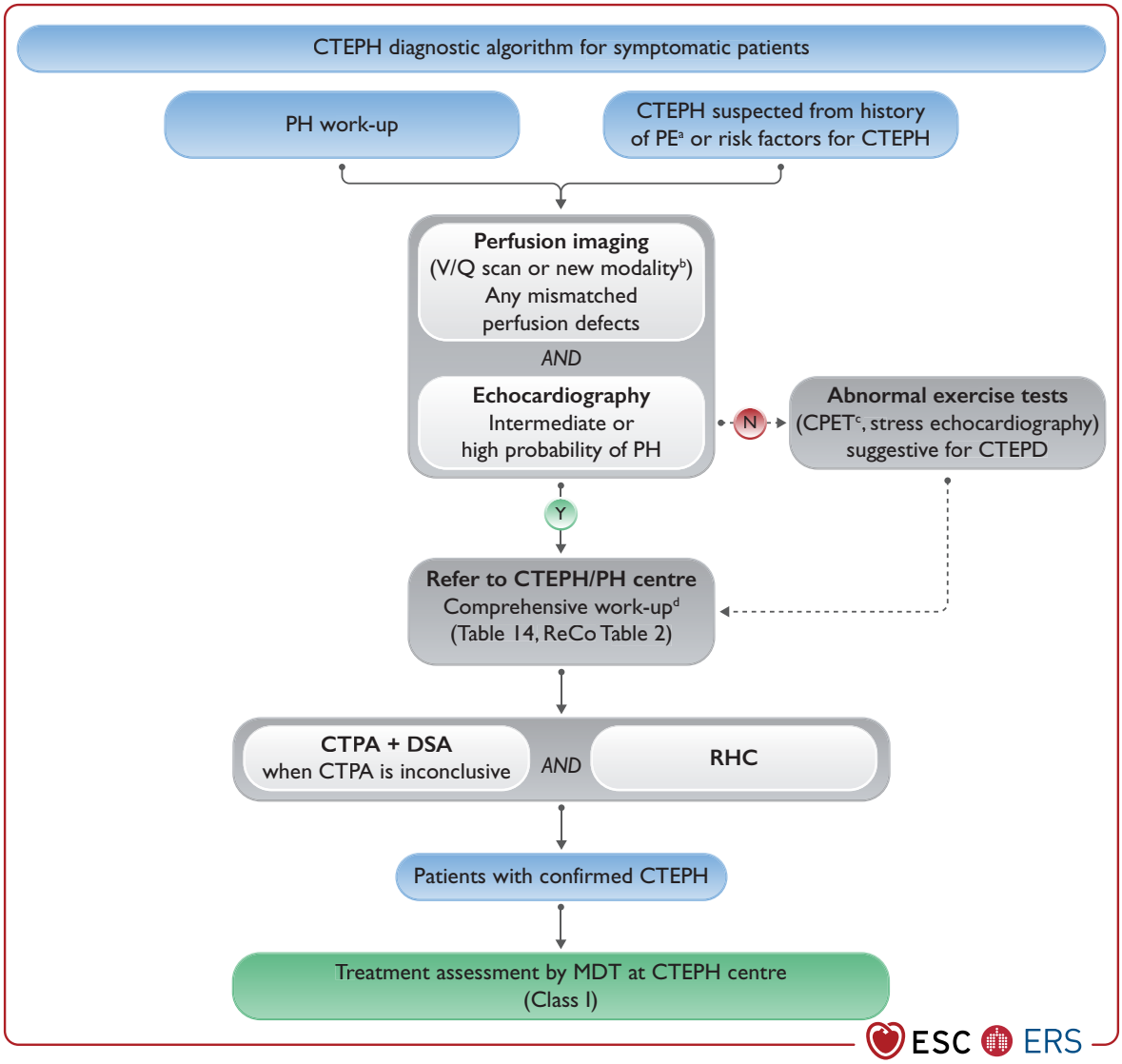


Figure 13 Diagnostic strategy in chronic thrombo-embolic pulmonary hypertension. CPET, cardiopulmonary exercise test; CTEPD, chronic thrombo-embolic pulmonary disease; CTEPH, chronic thrombo-embolic pulmonary hypertension; CTPA, computed tomography pulmonary angiography; DECT, dual-energy computed tomography; DSA, digital subtraction angiography; MDT, multidisciplinary team; MRI, magnetic resonance imaging; N, no; PE, pulmonary embolism; $P_{ET}CO_2$, end-tidal partial pressure of carbon dioxide; PH, pulmonary hypertension; ReCo, recommendation; RHC, right heart catheterization; sPAP, systolic pulmonary arterial pressure; V/Q, ventilation/perfusion; VE/VCO_2, ventilatory equivalents for carbon dioxide; VO_2/HR, oxygen pulse; VO_2, oxygen uptake; Y, yes. [a]CTEPH suspected from history of PE, including elevated sPAP on echocardiography and signs suggesting CTEPH on CTPA performed at the time of the acute PE (*Section 5.1.7*). [b]Alternative perfusion imaging techniques—such as iodine subtraction mapping, DECT, and MRI perfusion—are currently under evaluation. [c]Typical pattern, including low $P_{ET}CO_2$, high VE/VCO_2, low VO_2/HR, and low peak VO_2 (*Section 5.1.11*). [d]Comprehensive work-up after 3 months of therapeutic anticoagulation or sooner in unstable or rapidly deteriorating patients. Ideally, CTPA, DSA, and RHC are performed in CTEPH centres, but they are sometimes performed in PH centres, depending on the country and organization.

considered to have CTEPD with or without PH; CTEPH remains the preferred term for patients with PH, as defined in *Section 3.1* (*Table 5*).[54] Chronic thrombo-embolic pulmonary disease describes symptomatic patients with mismatched perfusion defects on V/Q scan and with signs of chronic, organized, fibrotic clots on CTPA or DSA, such as ring-like stenoses, webs/slits, and chronic total occlusions (pouch lesions or tapered lesions), after at least 3 months of therapeutic anticoagulation. Pulmonary hypertension in this setting is not only a consequence of PA obstruction by organized fibrotic clots but can also be related to the associated microvasculopathy. In those patients without PH at rest, breathlessness could be due to exercise PH (see definition in *Section 3.1*, *Table 5*) and/or increased dead space ventilation.[54] Excluding ventilatory limitation, deconditioning and psychogenic hyperventilation syndrome by CPET and LV myocardial or valvular disease by echocardiography is of upmost importance when making therapeutic decisions in patients with CTEPD without PH.

10.1. Diagnosis

Chronic thrombo-embolic pulmonary hypertension is a common and important cause of PH, with a distinct management strategy. Thus, the possibility of CTEPH should be carefully considered in all patients with PH (*Figure 13*). In the context of acute PE, CTEPH

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should be considered: (1) if radiological signs (detailed in *Section 5.1.7*) suggest CTEPH on the CTPA performed to diagnose PE,[112] and/or if estimated sPAP is >60 mmHg[112] on echocardiogram; (2) when dyspnoea or functional limitations persist in the clinical course post-PE;[754] and (3) in asymptomatic patients with risk factors for CTEPH or a high CTEPH prediction score.[755] Clinical conditions such as permanent intravascular devices (pacemaker, long-term central lines, ventriculoatrial shunts), inflammatory bowel diseases, essential thrombocythaemia, polycythaemia vera, splenectomy, antiphospholipid syndrome, high-dose thyroid hormone replacement, and malignancy are risk factors for CTEPH.[54,103,756]

Alternative causes of PA obstructions (also included in group 4 of the PH classification)—including PA sarcomas, other malignant tumours (e.g. renal carcinoma, uterine carcinoma, and germ-cell tumours of the testis), non-malignant tumours (e.g. uterine leiomyoma), arteriitis without CTD, congenital or acquired PA stenoses, parasites (hydatid cyst), and foreign-body embolism—have to be considered in the differential diagnosis of CTEPD.[757] They can be explored by specific additional imaging such as ^{18}F-2-fluoro-2-deoxy-D-glucose-positron emission tomography (PET) scan, which can provide additional information when PA sarcoma is suspected.[758]

Ventilation/perfusion scintigraphy[207] remains the most effective tool in excluding CTEPD. Alternative perfusion imaging techniques—such as iodine subtraction mapping, DECT, and MRI perfusion—have numerous theoretical advantages over V/Q but are more technically challenging and expensive, have limited availability, and currently lack multicentre validation.

Computed tomography pulmonary angiography with bi-planar reconstruction is broadly used for diagnosing CTEPD and assessing operability, but a negative CTPA, even if high quality, does not exclude CTEPD, as distal disease can be missed. Digital subtraction angiography is still used to assess treatment options when CTPA is inconclusive. Selective segmental angiography, cone-beam CT, and area detector CT allow for more accurate visualization of subsegmental vasculature and are useful for procedural guidance for BPA. The benefits of the new technologies require validating in prospective trials before being recommended for routine clinical use; a large, European, multicentre study is currently ongoing.[759]

10.2. Therapy

The CTEPH treatment algorithm includes a multimodal approach of combinations of pulmonary endarterectomy (PEA), BPA, and medical therapies to target the mixed anatomical lesions: proximal, distal, and microvasculopathy, respectively (*Figures 14* and *15*).

General measures recommended for PAH also apply to CTEPH, including supervised exercise training, which is effective and safe in inoperable CTEPH patients,[760] as well as early after PEA.[761]

Lifelong therapeutic anticoagulation is recommended for patients with CTEPH, as recurrent pulmonary thrombo-embolism accompanied by insufficient clot resolution are key pathophysiological features of this disease. There are no RCTs in CTEPH with any of the approved anticoagulants; however, despite this lack of evidence, VKAs are recommended by experts, and are most widely used as background therapy for patients with CTEPH. More recently, NOACs have more frequently been used as alternatives to VKAs, again, lacking evidence from RCTs. A retrospective case series from the UK and a multicentre prospective registry (EXPERT) showed comparable bleeding rates for VKAs and NOACs in CTEPH, but recurrent venous thrombo-embolism rates were higher in those receiving NOACs.[762,763] In patients with antiphospholipid syndrome (10% of the CTEPH population), VKAs are recommended.[103,764,765] Screening for antiphospholipid syndrome should be performed at CTEPH diagnosis. In the absence of any evidence in favour or against prolonged anticoagulation in patients with CTEPD without PH, long-term anticoagulant therapy is based on individual decision-making. It is recommended when the risk of PE recurrence is intermediate or high, thereby following the 2019 ESC/ERS Guidelines for the diagnosis and management of acute pulmonary embolism (*Table 11*).[103]

10.2.1. Surgical treatment

Surgical PEA is the treatment of choice for patients with accessible PA lesions.[102] As surgery may normalize pulmonary haemodynamics (65% decrease in PVR)[766] and functional capacity, an expert multidisciplinary team including an experienced PEA surgeon (on-site or closely collaborating) is mandatory for evaluating operability and deciding final treatment.[102]

Operability is based on team experience, accessibility of PA lesions, correlation between severity of PH and degree of PA obstructions, and comorbidities.[767] The surgical technique is complex but well standardized with >30 years of experience. It consists of a complete bilateral endarterectomy of the PAs down to segmental and subsegmental levels in phases of deep hypothermic circulatory arrest (*Figure 15*).[767,768] In CTEPH centres, surgical outcomes are favourable, with peri-operative mortality rates <2.5% due to improved management of cardiac and pulmonary complications and well-established use of ECMO.[768] Post-operative PH is frequently observed (~25%),[766] but long-term outcomes after PEA surgery are excellent regarding survival (averaging 90% at 3 years) and quality of life,[769–771] even in patients with distal PA obstructions.[772] On the other hand, patients with proximal operable disease declining surgery have a poor long-term outcome, with a 5 year survival of 53% compared with 83% in patients undergoing PEA.[773] Therefore, PEA should be offered to all operable patients with a favourable risk:benefit ratio, ideally during a personal consultation between the patient and the PEA surgeon.[102]

Selected symptomatic patients with CTEPD without PH can be successfully treated by PEA, with clinical and haemodynamic improvements at rest and exercise.[135,774] Those patients would require careful discussion to balance risk and benefit.

10.2.2. Medical therapy

To manage the microvascular component of CTEPH (*Figure 15*), medical therapies have been used off-label based on uncontrolled studies and/or regional approvals. Meanwhile, three RCTs have successfully been conducted. The first phase 3 RCT investigated the efficacy of riociguat in patients with inoperable CTEPH or those with persistent/recurrent PH after PEA.[775] Riociguat, after 16 weeks of therapy, improved 6MWD and reduced PVR by 31% compared with placebo, and is approved for this indication. Treprostinil s.c. was investigated in a phase 3 RCT, which showed improved 6MWD at week 24 in patients with inoperable CTEPH or those with persistent/recurrent PH after PEA receiving a high dose compared with a low dose;[776] s.c. treprostinil is approved for this indication. In a phase 2 study including only patients with inoperable CTEPH, macitentan 10 mg improved PVR and 6MWD vs. placebo

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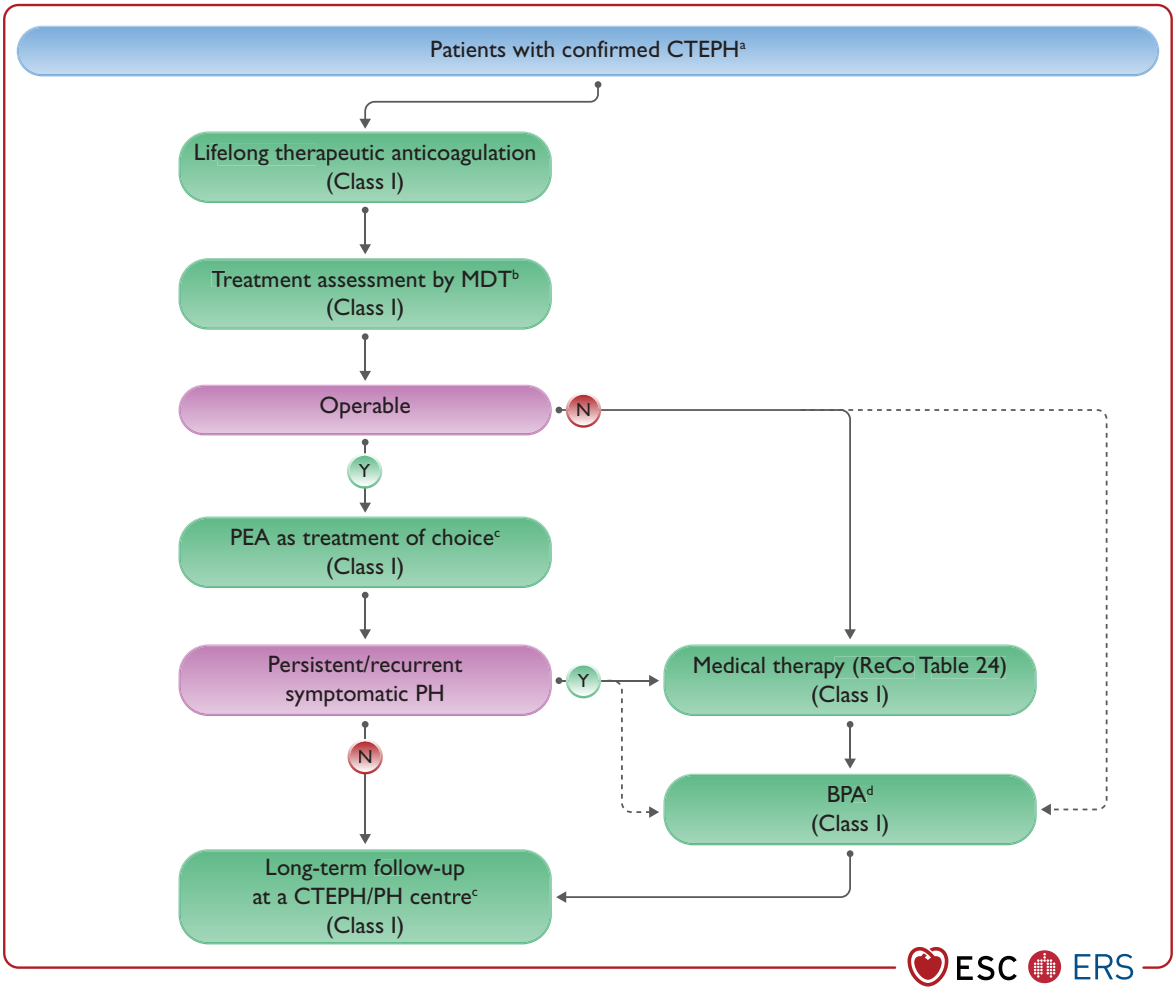


Figure 14 Management strategy in chronic thrombo-embolic pulmonary hypertension. BPA, balloon pulmonary angioplasty; CTEPD, chronic thrombo-embolic pulmonary disease; CTEPH, chronic thrombo-embolic pulmonary hypertension; MDT, multidisciplinary team; N, no; PAH, pulmonary arterial hypertension; PEA, pulmonary endarterectomy; PH, pulmonary hypertension; PVR, pulmonary vascular resistance; ReCo, recommendation; WU, Wood units; Y, yes. [a]Selected symptomatic patients with CTEPD without PH can also be treated by PEA and BPA. [b]MDT meeting can be virtual. [c]Treatment assessment may differ, depending on the level of expertise in PEA and BPA. [d]For inoperable patients with PVR >4 WU, medical therapy should be considered prior to BPA; there are limited data on BPA as first-line therapy.

at 16 and 24 weeks, respectively.[777] A phase 3 RCT is ongoing to evaluate the safety and efficacy of macitentan 75 mg in inoperable or persistent/recurrent CTEPH (NCT04271475).

Other medical therapies—PDE5is (e.g. sildenafil) and ERAs (e.g. bosentan)—have been used off-label, as their efficacy in inoperable CTEPH has not been proven by RCTs or registry data.[769,778,779] However, oral combination therapy, including PDE5is and ERAs, is common practice in patients with CTEPH with severe haemodynamic compromise.[780]

10.2.3. Interventional treatment

Balloon pulmonary angioplasty (*Figure 15*) has become an established treatment for selected patients with inoperable CTEPH or persistent/recurrent PH after PEA, improving haemodynamics (PVR decrease 49–66%), right heart function, and exercise capacity.[781–794] Long-term outcomes are promising, but evidence is still scarce.[795]

A staged interventional procedure with a limited number of dilated PA segments per session is preferred.[102,788] The number of sessions needed and haemodynamic results are dependent on experience.[781] While BPA is effective, it is associated with serious complications, which may be fatal. Procedural and post-interventional complications include vascular injury due to wire perforation, and lung injury with haemoptysis and/or hypoxia.[102,781,796,797] As with all interventional procedures, a significant learning curve has been shown, with reducing complication rates over time;[781] therefore, this procedure should be performed in high-volume CTEPH centres. As the rates of interventional complications can be reduced by medical pre-treatment, patients with a PVR >4 WU should be treated before BPA (*Figure 15*).[798]

Selected symptomatic patients with CTEPD without PH and segmental/subsegmental lesions can successfully be treated by BPA, with clinical and haemodynamic improvements at rest and exercise.[799]

Preliminary data on PADN point towards improved exercise capacity and pulmonary haemodynamics in patients with persistent PH after PEA;[800] further confirmation is being awaited.

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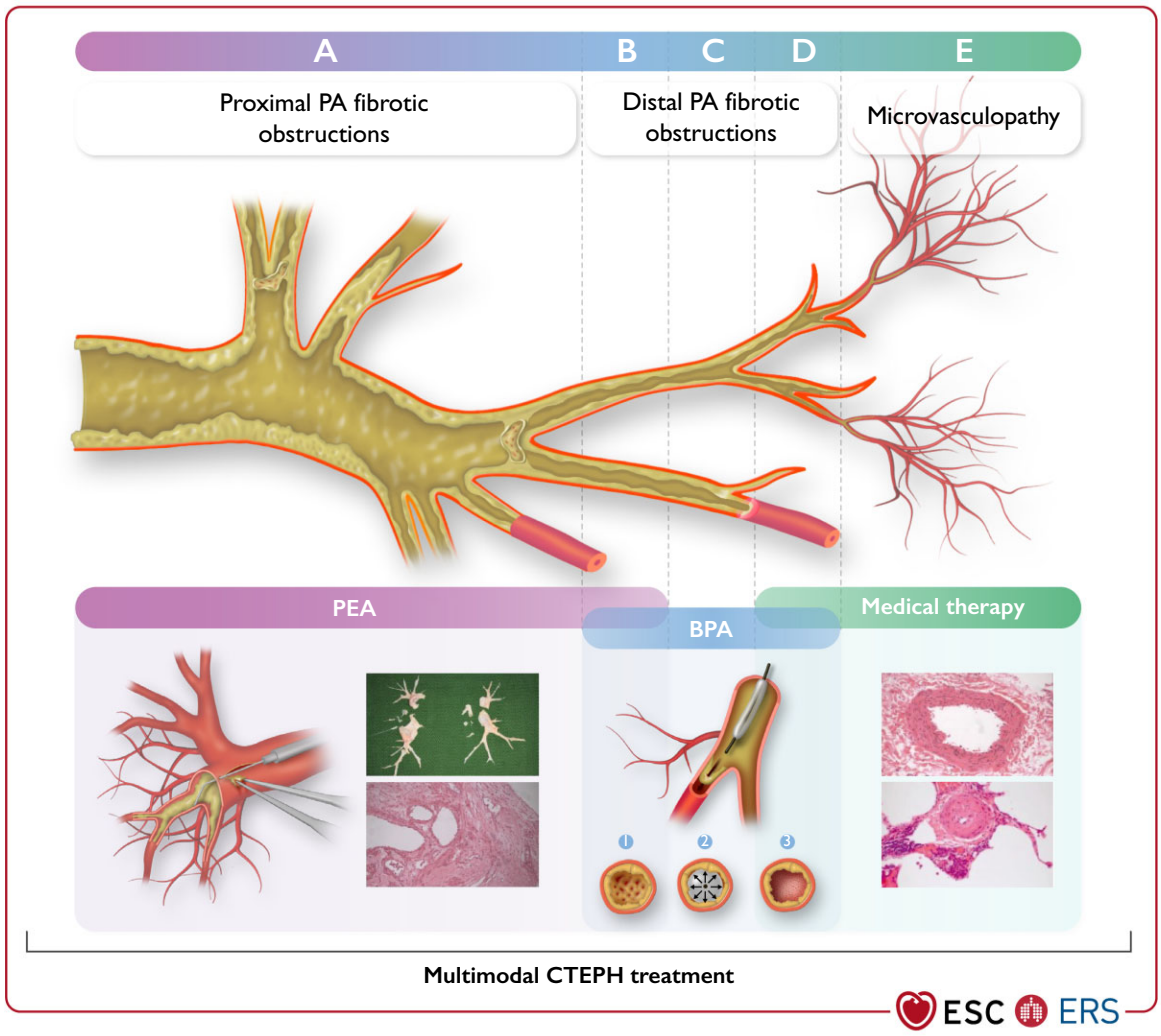


Figure 15 Overlap in treatments/multimodality approaches in chronic thrombo-embolic pulmonary hypertension. BPA, balloon pulmonary angioplasty; CTEPH, chronic thrombo-embolic pulmonary hypertension; PA, pulmonary artery; PEA, pulmonary endarterectomy. Top panels: (*A*) Proximal PA fibrotic obstructions (vessel diameter 10–40 mm). (*B*) Distal segmental and subsegmental PA fibrotic obstruction potentially suitable for both PEA and BPA interventions (vessel diameter 2–10 mm). (*C*) Distal subsegmental PA fibrotic obstructions form a web-lesion in a subsegmental branch of the PA suitable for BPA interventions (vessel diameter 0.5–5 mm). (*D*) Distal subsegmental PA fibrotic obstructions form web-like lesions, which might be accompanied by microvasculopathy (vessel diameter <0.5 mm). (*E*) Microvasculopathy (vessel diameter <0.05 mm) treated with medical therapy. Bottom panels: (*A*) bottom left: PEA; vessel diameter (0.2–3 cm). The right PA is opened and the suction dissector is introduced between the artery wall and fibrosis. Following the inside of the artery down to segmental and subsegmental levels, the fibrotic material is subsequently freed from the wall and removed with forceps. (*A*) bottom right: PEA specimen with 'tails' to subsegmental branches of the PA; cross-section of partially organized and permeabilized thrombotic lesion of the large PA dissected during PEA. (*B*, *C*, *D*) The wire is introduced between the fibrotic material (*1*), then the balloon is inflated, leading to a rupture of the web (*2*). Fibrotic material is connected to the vessel wall (*3*). (*E*) Small muscular PA displaying eccentric intimal fibrosis involving intimal thickening and proliferation—target for medical therapies.

10.2.4. Multimodal treatment

Multimodal therapy including surgery, medication, and intervention is offered to selected patients with CTEPH (*Figure 15*).[102]

Using medical therapy in patients with high pre-operative PVR to improve pulmonary haemodynamics before PEA is common practice but still controversial, as it is felt to delay timely surgical referral and therefore definitive treatment.[801–803]

A significant proportion of symptomatic patients may have persistent or recurrent PH following PEA, which may also benefit from medical and/or interventional therapies (*Figure 15*).[804–806] An mPAP ≥30 mmHg has been associated with initiation of medical therapies post-PEA, and an mPAP ≥38 mmHg and PVR ≥5 WU with worse long-term survival.[806]

Some patients with CTEPH may have mixed anatomical lesions, with surgically accessible lesions in one lung and inoperable lesions in the other lung. Such patients might benefit from a combined approach with BPA (prior to or at the same time as surgery) and PEA to decrease the surgical risk and improve the final result.[807]

The recommendations on BPA and medical therapy in patients with inoperable CTEPH have been established based on key narrative question 8 (Supplementary Data, *Section 10.1*).

The recommendation on the use of medical therapy before interventional therapy in patients with CTEPH who are considered inoperable but candidates for BPA is based on PICO question IV (Supplementary Data, *Section 10.2*). The included evidence suggests

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that pre-treatment improves pulmonary haemodynamics and safety of the procedure. This is confirmed by the clinical experience of Task Force members. However, due to the low certainty of the evidence, the recommendation is conditional.

10.2.5. Follow-up

Regardless of the result of PEA/BPA, patients should be regularly followed-up, including invasive assessment with RHC 3–6 months after intervention, allowing for consideration of a multimodal treatment approach. After successful treatment, yearly non-invasive follow-up, including echocardiography and an evaluation of exercise capacity, is indicated because recurrent PH has been described (*Figure 14*).[806]

Risk assessment with either the ESC/ERS or REVEAL risk score developed for PAH has been validated in medically treated patients with CTEPH,[300,808,809] but it is unknown if its use has any therapeutic implication or affects outcome.

There are no data or consensus on what is the therapeutic target after PEA/BPA or medical therapy in CTEPH. Most experts accept achieving a good functional class (WHO-FC I–II) and/or normalization or near normalization of haemodynamics at rest, obtained at RHC 3–6 months post-procedure (PEA or last BPA), and improvement in quality of life.

10.3. Chronic thrombo-embolic pulmonary hypertension team and experience criteria

To optimize patients' outcomes, CTEPH centres should fulfil criteria for a PH centre (*Section 12*) and have a CTEPH multidisciplinary team consisting of a PEA surgeon, BPA interventionist, PH specialist, and thoracic radiologist, trained in high-volume PEA and/or BPA centres. The team should meet regularly to review new referrals and post-treatment follow-up cases. Ideally, CTEPH centres should have PEA activities (>50/year)[810] and BPAs (>30 patients/year or >100 procedures/year),[781] as these figures have been associated with better outcome. The CTEPH centres should also manage medically treated patients. Based on regional requirements, these numbers may be adjusted for the country's population, ideally concentrating care and expertise in high-volume centres.

Recommendation Table 24 — Recommendations for chronic thrombo-embolic pulmonary hypertension and chronic thrombo-embolic pulmonary disease without pulmonary hypertension

Recommendation Table 24A

Recommendations	Class[a]	Level[b]
CTEPH		
Lifelong, therapeutic doses of anticoagulation are recommended in all patients with CTEPH[762]	I	C
Antiphospholipid syndrome testing is recommended in patients with CTEPH	I	C
In patients with CTEPH and antiphospholipid syndrome, anticoagulation with VKAs is recommended[103,764,765]	I	C
It is recommended that all patients with CTEPH are reviewed by a CTEPH team for the assessment of multimodality management[54]	I	C
PEA is recommended as the treatment of choice for patients with CTEPH and fibrotic obstructions within pulmonary arteries accessible by surgery[54,102]	I	B
BPA is recommended in patients who are technically inoperable or have residual PH after PEA and distal obstructions amenable to BPA[54,102,783,784,789,793,798,811]	I	B
Riociguat is recommended for symptomatic patients with inoperable CTEPH or persistent/recurrent PH after PEA[775]	I	B
Long-term follow-up is recommended after PEA and BPA, as well as for patients with CTEPH established on medical therapy[782,805,806,812]	I	C
A multimodality approach should be considered for patients with persistent PH after PEA and for patients with inoperable CTEPH[804,805,812]	IIa	C
Treprostinil s.c. may be considered in patients in WHO-FC III–IV who have inoperable CTEPH or persistent/recurrent PH after PEA[776]	IIb	B
Off-label use of drugs approved for PAH may be considered in symptomatic patients who have inoperable CTEPH[55,777–779,801]	IIb	B
In patients with inoperable CTEPH, a combination of sGC stimulator/PDE5i, ERA,[777] or parenteral prostacyclin analogues[776] may be considered	IIb	C
BPA may be considered for technically operable patients with a high proportion of distal disease and an unfavourable risk:benefit ratio for PEA	IIb	C
CTEPD without PH		
In patients with CTEPD without PH, long-term anticoagulant therapy should be considered on an individual basis[c]	IIa	C
PEA or BPA should be considered in selected symptomatic patients with CTEPD without PH	IIa	C

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Recommendation Table 24B

Recommendations	GRADE		Class[a]	Level[b]
	Quality of evidence	Strength of recommendation		
In patients with CTEPH who are candidates for BPA, medical therapy should be considered prior to the intervention[798]	Very low	Conditional	IIa	B

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BPA, balloon pulmonary angioplasty; CTEPD, chronic thrombo-embolic pulmonary disease; CTEPH, chronic thrombo-embolic pulmonary hypertension; ERA, endothelin receptor antagonist; PAH, pulmonary arterial hypertension; PDE5i, phosphodiesterase 5 inhibitor; PEA, pulmonary endarterectomy; PE, pulmonary embolism; PH, pulmonary hypertension; s.c., subcutaneous; sGC, soluble guanylate cyclase; VKA, vitamin K antagonist; WHO-FC, World Health Organization functional class.
[a]Class of recommendation.
[b]Level of evidence.
[c]Long-term anticoagulant therapy is recommended when the risk of PE recurrence is intermediate or high,[103] or when there is no history of venous thrombo-embolism.

11. Pulmonary hypertension with unclear and/or multifactorial mechanisms (group 5)

Pulmonary hypertension with unclear and/or multifactorial mechanisms (*Table 24*) includes several conditions that may be complicated by complex and sometimes overlapping pulmonary vascular involvement. Although group 5 PH represents less-studied forms of PH, it constitutes a significant part of the worldwide burden of PH.[1] Group 5 PH includes: haematological disorders, such as SCD and chronic myeloproliferative neoplasms; systemic disorders, such as sarcoidosis; metabolic diseases, such as glycogen storage disease; and others, such as chronic renal failure, pulmonary tumour thrombotic microangiopathy, and fibrosing mediastinitis. A common feature of these diseases is that the mechanisms of PH are poorly understood and contributing factors may include, alone or in combination: hypoxic pulmonary vasoconstriction, pulmonary vascular remodelling, thrombosis, fibrotic destruction and/or extrinsic compression of pulmonary vasculature, pulmonary vasculitis, high-output cardiac failure, and left HF. These patients need careful assessment and treatment should be directed to the underlying condition.

Table 24 Pulmonary hypertension with unclear and/or multifactorial mechanisms

Disorders associated with pulmonary hypertension	
1 Haematological disorders	Inherited and acquired chronic haemolytic anaemia • Sickle cell disease • β-thalassaemia • Spherocytosis • Stomatocytosis • Autoimmune disorders Chronic myeloproliferative disorders • Chronic myelogenous leukaemia • Polycythaemia vera • Idiopathic myelofibrosis • Essential thrombocytopenia • Others
2 Systemic disorders	Sarcoidosis Pulmonary Langerhans's cell histiocytosis Neurofibromatosis type 1
3 Metabolic disorders	Glycogen storage disease Gaucher disease
4 Chronic renal failure with/without haemodialysis	
5 Pulmonary tumour thrombotic microangiopathy	
6 Fibrosis mediastinitis	

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11.1. Haematological disorders

In haemoglobinopathies and chronic haemolytic anaemias, including SCD, PH has emerged as a major cause of morbidity and mortality. The prevalence of PH confirmed by RHC was 6–10% in studies of adult patients with stable SCD.[93,94,813] Patients with SCD with pre-capillary PH are more commonly homozygous for haemoglobin S, while some have S-β0 thalassaemia (S-β0 thal) or haemoglobin SCD.[814] Thrombotic lesions are a major component of PH related to SCD, more frequently in haemoglobin SCD.[814] Patients with PH and SCD should be followed by multidisciplinary SCD and PH teams, since treatment of the anaemia is a key part of management.[814] There is a lack of data to support the use of PAH drugs in patients with SCD-associated PH. In a study in patients with SCD with TRV ≥2.7 m/s and a 6MWD of 150–500 m, sildenafil showed no treatment effect on 6MWD, TRV, or NT-proBNP, but appeared to increase hospitalization rates for pain.[815] Preliminary evidence supports the short- and long-term benefits of chronic blood-exchange transfusions in patients with pre-capillary PH complicating SCD.[816] Pre-capillary PH complicating SCD has an important impact on survival, with an overall death rate of 2.0–5.3% in different populations with similar follow-up (26 months and 18 months, respectively).[94,817] In β-thalassaemia, invasive haemodynamic evaluation confirmed pre-capillary PH in 2.1% of cases, while a post-capillary profile was found in 0.3%.[818] Potential treatment strategies are awaiting an enhanced understanding of the

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pathophysiological mechanisms. In spherocytosis, splenectomy is a risk factor for CTEPH.[819]

Multiple causes of PH have been described in patients with chronic myeloproliferative disorders.[820] In chronic myelogenous leukaemia, spleen enlargement and anaemia can give rise to hyperkinetic syndrome. Hepatosplenic enlargement can also cause PoPH. Cases of potentially reversible DPAH have been described with dasatinib, bosutinib, and ponatinib. In polycythaemia vera and essential thrombocythaemia, there is an increased risk of venous thrombo-embolic disease and CTEPH; moreover, a blood clot within the hepatic veins can lead to Budd–Chiari syndrome and subsequent PoPH. Pulmonary extramedullary haematopoiesis complicating idiopathic or secondary myelofibrosis may also contribute to dyspnoea and PH.

Group 5 PH may be described in other haematological disorders, such as common variable immunodeficiency; immunoglobulin G4 (IgG4)-related disease; Castleman disease; and polyneuropathy, organomegaly, endocrinopathy, monoclonal immunoglobulin, skin changes (POEMS) syndrome.[821–823]

11.2. Systemic disorders

The reported prevalence of PH in patients with sarcoidosis is 6–20%.[824] The causes are multifactorial, including fibrosing lung disease, granulomata in the PAs and/or pulmonary veins, fibrosing mediastinitis and/or extrinsic compression by lymph nodes, pulmonary vasculitis, CTEPH, and PoPH.[58,825] It is associated with significant morbidity and increased mortality compared with sarcoidosis without PH.[58,825] In a registry, factors independently associated with outcomes included physiological (forced expiratory volume in 1 s/FVC ratio and DLCO) and functional (6MWD) parameters.[58] In a large study of severe, sarcoidosis-associated PH, PAH drugs improved short-term pulmonary haemodynamics without improving 6MWD.[59] Small RCTs have suggested efficacy of PAH drugs in these patients, which requires confirmation in larger studies.[826] Corticosteroids or immunosuppressive therapy may improve haemodynamics in selected patients with active granulomatous disease. Of note, when pulmonary vascular compression is suspected (fibrosing mediastinitis and/or extrinsic compression by lymph nodes), results from pulmonary angiography and PET scans provide additional information justifying endovascular and/or anti-inflammatory approaches. Long-term survival remains poor in sarcoidosis-associated PH, which makes LTx a reasonable option for selected severe cases.

In pulmonary Langerhans's cell histiocytosis, diminished exercise capacity does not appear to be due to ventilatory limitation but may be related to pulmonary vascular dysfunction. In 29 patients with PH associated with pulmonary Langerhans's cell histiocytosis, PAH drugs improved haemodynamics without worsening oxygen levels.[827]

Pulmonary hypertension associated with neurofibromatosis type 1 is a rare but severe complication characterized by female predominance (female/male ratio 3.9:1).[828] Specific pulmonary vascular involvement exists in these patients, and despite a potential short-term benefit of PAH drugs, prognosis remains poor, and LTx should be considered in selected patients with severe disease. In the presence of dyspnoea, screening for ILD by non-contrast CT and for PH by echocardiography is required.[828]

11.3. Metabolic disorders

Glycogen storage diseases are caused by genetic alterations of glycogen metabolism, and PH case reports have been related to glycogen storage disease type 1 and 2.[829] The occurrence of PH has predominantly been described in glycogen storage disease type 1, where it may partly be due to vasoconstrictive amines such as serotonin. Drugs for PAH have been used in some cases.[830]

Untreated patients with Gaucher disease may develop PH, which is caused by a combination of factors, including asplenia, plugging of the vasculature by abnormal macrophages, and pulmonary vascular remodelling. Treatment with enzyme-replacement therapy may improve PH.

11.4. Chronic kidney failure

Although commonly recognized in chronic renal failure, the pathogenesis of PH remains poorly understood and PH is observed in patients prior to and while receiving different dialysis modalities.[831] A recent RHC study of 3504 patients with chronic kidney disease found that CpcPH was the most common phenotype, and the phenotype with the highest mortality.[832] Post-capillary PH has been described in 65% of patients receiving haemodialysis and 71% of patients without kidney replacement.[833]

11.5. Pulmonary tumour thrombotic microangiopathy

Pulmonary tumour thrombotic microangiopathy describes tumour-cell microemboli with occlusive fibrointimal remodelling in small PAs, pulmonary veins, and lymphatics. It is a rare cause of PH, which arises due to multiple mechanisms, but probably remains under-diagnosed, as evidenced by autopsy findings.[834] The disorder is associated with carcinomas, notably gastric carcinoma. Progressive vessel occlusion ultimately results in PH, which is often severe, of sudden onset, rapidly progressive, and accompanied by progressive hypoxaemia. Chest CT may show patchy ground-glass and septal markings (masquerading as PVOD).

11.6. Fibrosing mediastinitis

Fibrosing mediastinitis is caused by fibrous tissue proliferating in the mediastinum, encasing mediastinal viscera and compressing mediastinal bronchovascular structures.[835] Pre- or post-capillary PH can complicate the course of fibrosing mediastinitis due to extrinsic compression of the PAs and/or pulmonary veins. Fibrosing mediastinitis can be idiopathic or caused by irradiation, infection (tuberculosis, histoplasmosis), and systemic diseases, such as sarcoidosis and IgG4-related disease, a fibroinflammatory disease characterized by elevated serum IgG4 levels with infiltration of IgG4+ plasma cells and severe fibrosis in affected tissues.[821] Treatment should be directed to the underlying condition. No clear clinical improvement has been described with PAH drugs. Surgical and endovascular procedures have been proposed to de-obstruct or bypass the arterial and/or venous compressions.

In the absence of positive RCTs studying the use of PAH drugs for treating group 5 PH, treating the underlying disorder remains the standard of care.[836] Importantly, some of the diseases described in *Table 24* may have a pulmonary venous component that could be made worse with PAH drugs, implying that off-label use of drugs approved for PAH should be considered with great caution, if at all.

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Placebo-controlled, randomized trials are currently recruiting in well-phenotyped subgroups of PH with unclear and/or multifactorial mechanisms, such as sarcoidosis-associated PH.

12. Definition of a pulmonary hypertension centre

While PH is not an uncommon condition, severe forms of PH, especially PAH and CTEPH, require highly specialized management. Since medical centres with multidisciplinary teams and a high volume of patients generally offer best standard of care, which translates into better clinical outcomes, establishing PH centres is clinically and economically highly desirable and is supported by patient organizations and scientific societies. The purpose of a PH centre is to: receive new referrals; assess and investigate the cause of PH; carefully phenotype and routinely manage patients with medical, interventional, and surgical approaches; work closely with other health care providers to achieve the best outcomes for patients; undertake audits (reporting patient case mix and quality indicators); and be involved in clinical and translational research, and education. The requirements—comprising definition, multidisciplinary structure, number of cases, procedures, and staffing levels, as well as the skills and resources needed in a PH referral centre—are described below and in *Figure 16*. Criteria for paediatric and CTEPH centres are described elsewhere (*Sections 7.8.3* and *10.3*, respectively).

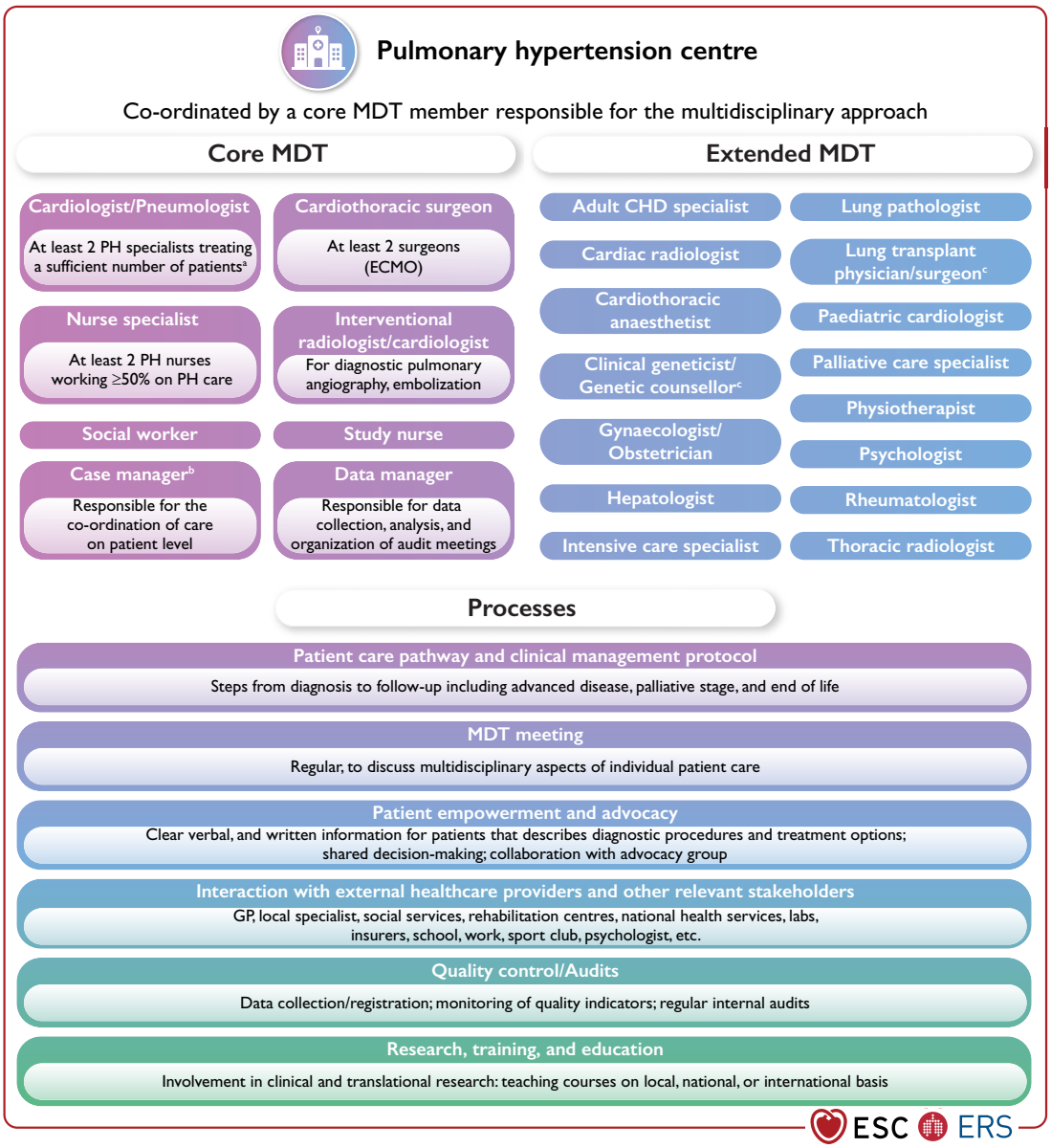


Figure 16 Pulmonary hypertension centre schematic. CHD, congenital heart disease; ECMO, extracorporeal membrane oxygenation; GP, general practitioner; MDT, multidisciplinary team; PH, pulmonary hypertension. [a]Number adapted according to specific country characteristics. [b]Case manager can be a nurse specialist, social worker, physiotherapist, or administrative assistant in function of the centre organization. [c]Can be located in partner centres. Adapted from Biganzoli *et al.*[846]

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12.1. Facilities and skills required for a pulmonary hypertension centre

Pulmonary hypertension centres care for a sufficient number of patients on PH therapy, as well as new referrals, to warrant this status. According to the 2015 ESC/ERS Guidelines for the diagnosis and treatment of PH and the European Reference Network on rare respiratory diseases (ERN-LUNG) competency requirements, the ideal number of patients seen by an adult centre each year is no fewer than 200, of which at least half have a final diagnosis of PAH; a PH centre follows at least 50 patients with PAH or CTEPH and receives at least two new referrals per month with documented PAH or CTEPH.[25,26,837–839] These numbers can be adapted according to specific country characteristics (small population, large geographical area) provided that strong working collaborations are established with high-volume centres. This is currently facilitated by the availability of secure virtual platforms (e.g. ERN clinical patient management system).[840]

Proper training of staff members includes core competencies, such as those outlined in the ERS Pulmonary Vascular Diseases Continuing Professional Development framework,[841] and builds on entrustable professional activities, described in the ESC Core Curriculum.[842]

Clinical, laboratory, and imaging facilities include: a ward where health care providers have expertise in PH; a specialist outpatient service; an intermediate/ICU; 24/7 emergency care; an interventional radiology unit; diagnostic investigations, including echocardiography, CT scanning, nuclear medicine, MRI, exercise tests, and PFTs; a cardiac catheterization laboratory; access to genetic counselling and testing; and fast and easy access to cardiothoracic and vascular surgery. Key diagnostic procedures are performed in sufficient numbers to guarantee expertise (e.g. ERN-LUNG requirements).[837] In analogy with the 'advanced heart failure units',[843] PH centres offer the full range of PAH therapies available in their country (including i.v./s.c. prostacyclin derivatives) and have early referral protocols to CTEPH, LTx, and rehabilitation centres. Since evaluation and early availability of new drugs and techniques are critical, PH centres participate in collaborative clinical research.

Regular multidisciplinary team meetings, including core members and on-demand invited members (extended multidisciplinary team) as needed (*Figure 16*), are required to establish and adapt individual patient care pathways. Case management (co-ordination of individual patient pathways) should include administrative, social, and care support. Remote accessibility of the PH centre by phone, mail, or other is a vital part of the care. Strategies have to be implemented in order to improve health literacy and shared decision-making, with the support of dedicated patient decision tools. Transitioning from a paediatric PH centre to an adult PH centre requires adequate planning to prevent gaps in care. Involving national and/or international patient associations helps to design patient-centric care and to spread medical knowledge among patients and their carers.

Pulmonary hypertension centres should record patients' data using local, national, or international patient registries, and be able to report process indicators (compliance with diagnostic and treatment guidelines, including LTx) and outcome indicators, such as WHO-FC, exercise capacity, haemodynamics, quality of life, complications, and survival. They should undergo regular audits to assess the quality of delivered care.

Recommendation Table 25 — Recommendations for pulmonary hypertension centres

Recommendations	Class[a]	Level[b]
It is recommended that PH centres provide care by a multidisciplinary team (cardiologist, pneumologist, rheumatologist, nurse specialist, radiologist, psychological and social work support, and appropriate on-call expertise)	I	C
It is recommended that PH centres have direct links and quick referral patterns to other services (such as genetic counselling, PEA/BPA, LTx, and adult congenital heart disease service)	I	C
It is recommended that PH centres maintain a patient registry	I	C
It is recommended that PH centres collaborate with patient associations	I	C
Accreditation of the PH centres should be considered (e.g. https://ec.europa.eu/health/ern/assessment_en)	IIa	C
PH centres' participation in collaborative clinical research should be considered	IIa	C
PH centres should follow-up a sufficient number of patients to maintain expertise (at least 50 patients with PAH or CTEPH and at least two new referrals per month with documented PAH or CTEPH), and consider establishing collaborations with high-volume centres	IIa	C

BPA, balloon pulmonary angioplasty; CTEPH, chronic thrombo-embolic pulmonary hypertension; LTx, lung transplantation; PAH, pulmonary arterial hypertension; PEA, pulmonary endarterectomy; PH, pulmonary hypertension.
[a]Class of recommendation.
[b]Level of evidence.

12.2. European Reference Network

In 2017, the European Commission launched European Reference Networks (ERNs) for rare diseases that included the ERN-LUNG with a PH core network. European Reference Networks are patient-centred networks of commissioned centres offering guidance and cross-border best standard of care in the European Union. The PH network includes over 20 full members, contributing each year ~1500 new patients with PAH or CTEPH.[844] It also includes UK supporting centres, and affiliated partners (who do not necessarily have to fulfil the minimum competency criteria of the ERN-LUNG PH network). The ERN-LUNG requires and monitors standards for these centres.

12.3. Patient associations and patient empowerment

Pulmonary hypertension centres should inform patients about patient associations and encourage them to join such groups. Patient associations are a valuable resource for managing patients, as they provide educational and emotional support, and can have positive effects on coping, confidence, and outlook.[845] It is recommended that PH centres collaborate with patient associations on initiatives to

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empower patients and improve the patient experience, addressing issues such as health literacy, digital skills, healthy lifestyles, mental health, and self-management. Health care can be delivered more effectively and efficiently if patients are full partners in the process.

13. Key messages

(1) The haemodynamic definition of PH has been updated as mPAP >20 mmHg. The definition of PAH also implies a PVR >2 WU and PAWP ≤15 mmHg. These cut-off values better reflect the limits of normal ranges, but do not yet translate into new therapeutic recommendations, since the efficacy of PAH therapy in patients with PVD and an mPAP 21–24 mmHg and/or PVR 2–3 WU is still unknown.

(2) The main diagnostic algorithm for PH has been simplified following a three-step approach, from suspicion by first-line physicians, detection by echocardiography, and confirmation with RHC in PH centres. Warning signs associated with worse outcomes have been identified, which justify immediate referral and management in PH centres.

(3) Screening strategies for PAH in patients with SSc and in those at risk of HPAH are proposed based on the results of published cohort studies. Their implementation may shorten the time from symptom onset to diagnosis of PAH.

(4) An improved recognition of CT and echocardiographic signs of CTEPH at the time of an acute PE event, together with a systematic follow-up of patients with acute PE, as indicated in the 2019 ESC/ERS Guidelines for the diagnosis and management of acute pulmonary embolism, should help to remediate the underdiagnosis of CTEPH.

(5) The three-strata risk-stratification assessment in PAH has been refined after being validated in multiple registries. The MRI and echocardiographic criteria have been added to the ESC/ERS table, refining non-invasive evaluation at diagnosis.

(6) A four-strata risk stratification, dividing the large, intermediate-risk group into intermediate–low and intermediate–high risk, is proposed at follow-up.

(7) The treatment algorithm for PAH has been simplified, with a clear focus on risk assessment, cardiopulmonary comorbidities, and treatment goals. Initial combination therapy and treatment escalation at follow-up when appropriate are current standards.

(8) The Task Force has attempted to close the gap between paediatric and adult PAH care, with therapeutic and follow-up strategies based on risk stratification and treatment response, extrapolated from that in adults but adapted for age.

(9) The recommendations on sex-related issues in patients with PAH, including pregnancy, have been updated, with information and shared decision-making as key points.

(10) The recommendations for rehabilitation and exercise programmes in PH have been updated following the release of additional supportive evidence.

(11) For the first time, there is a recommendation for PH medical therapy in group 3 PH, based on a single positive RCT in patients with ILD.

(12) The concept of CTEPD with or without PH has been introduced, enabling further research on the natural history and management in the absence of PH.

(13) The treatment algorithm for CTEPH has been modified, including multimodal therapy with surgery, PH drugs, and BPA.

14. Gaps in evidence

14.1. Pulmonary arterial hypertension (group 1)

- The efficacy and safety of PAH drugs in group 1 patients with an mPAP 21–24 mmHg, PVR 2–3 WU, and exercise PH has to be established.
- The role of PAH drugs in different PAH subgroups, including schistosomiasis-associated PAH, needs to be explored.
- Risk-stratification assessment in PAH needs to be further prospectively validated through goal-orientated outcome studies, and optimized for patients with PAH and comorbidities.
- New PAH phenotypes observed in patients with significant cardiopulmonary comorbidities are common and should be the focus of more research.
- The importance of PAH patient phenotypes and the relevance of comorbidities on treatment goals and outcomes must be further evaluated.
- The impact of PAH therapies and treatment strategies on survival needs to be further assessed.
- Pulmonary arterial hypertension drugs targeting novel pathways are emerging and the impact of add-on use of this medication on outcomes has to be evaluated in RCTs.
- The role of RV imaging techniques (echocardiography, cMRI) in diagnosing and stratifying risk in PAH needs to be further studied. The proposed cut-off values for risk stratification need to be properly validated in multicentre studies.
- The role of CPET in the early diagnosis of PAH in populations at risk of developing PAH, and in assessing prognosis in PAH on top of clinical and haemodynamic data, needs further investigation.
- The role of exercise echocardiography and exercise RHC in patients at risk of developing PAH, with abnormal CPET but normal rest echocardiogram, also needs further evaluation.
- The use of mechanical circulatory support, particularly in reversible PH or in patients with advanced right HF with an exit strategy (such as LTx), has to be further studied.
- Differences in natural history and treatment response between adults and children should be further investigated.
- Further studies are needed on the effects of PADN in PAH and in other PH groups.
- The impact of centre volume, organization, and expertise on treatment outcome needs further investigation.

14.2. Pulmonary hypertension associated with left heart disease (group 2)

- The management of patients with group 2 PH needs further study with RCTs.
- Additional research is needed to facilitate non-invasive diagnosis of HFpEF-associated PH and distinguishing it from PAH.
- The role of fluid challenge and exercise testing to reveal left HF needs further validation.

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- Further studies focusing on PDE5is in patients with HFpEF and a CpcPH phenotype are needed and currently underway.
- The effects that new HF medication (ARNIs, SGLT-2is) has on PH, through reverse remodelling of the LV, need further investigation.

14.3. Pulmonary hypertension associated with lung diseases and/or hypoxia (group 3)

- The management of patients with group 3 PH has to be further studied in RCTs.
- Refining phenotypes will be crucial, as this will inform development of trials.
- Clinical relevance and therapeutic implications of severe PH in lung disease need to be investigated.
- Long-term data on the effects of inhaled treprostinil (and other PAH drugs) in patients with PH associated with lung disease are needed.
- The impact of the hypobaric and hypoxic environment of the >150 million people living at >2500 m altitude has to be clarified, and studies need to be performed to assess potential treatment strategies for PH.

14.4. Chronic thrombo-embolic pulmonary hypertension (group 4)

- The differentiation between acute and chronic PE in imaging (CTPA) has to be improved.
- In patients with suspected CTEPH, the diagnostic role of DECT or iodine subtraction mapping vs. V/Q lung scintigraphy has to be validated.
- The effect of drug therapy on the outcome of patients with CTEPH needs to be established.
- The treatment goals in patients with CTEPH have to be clarified, as it is still unclear if normalizing mPAP and PVR translates into improved outcomes.
- The role of BPA vs. PEA should be further clarified: which treatment in which patient? Are they equivalent for the treatment of segmental/subsegmental disease?
- In inoperable CTEPH or persistent/recurrent PH after PEA, the potential role of combination therapy of PH drugs must be assessed.
- The role of medical treatments as bridges to interventional and operative treatments needs to be formally tested.
- Randomized controlled trials are needed to discriminate the effects of PEA and early follow-up rehabilitation.
- The effect of PEA, BPA, and medical therapy on patients with CTEPD without PH is not established.

14.5. Pulmonary hypertension with unclear and/or multifactorial mechanisms (group 5)

- Further research needs to inform management of group 5 PH, such as SCD-associated PH and sarcoidosis-associated PH.

15. 'What to do' and 'What not to do' messages from the Guidelines

Recommendations	Class[a]	Level[b]
Recommendations for right heart catheterization and vasoreactivity testing		
Right heart catheterization		
It is recommended that RHC is performed to confirm the diagnosis of PH (especially PAH or CTEPH) and to support treatment decisions	I	B
In patients with suspected or known PH, it is recommended that RHC is performed in experienced centres	I	C
It is recommended that RHC comprises a complete set of haemodynamics and is performed following standardized protocols	I	C
Vasoreactivity testing		
Vasoreactivity testing is recommended in patients with I/H/DPAH to detect those who can be treated with high doses of a CCB	I	B
It is recommended that vasoreactivity testing is performed at PH centres	I	C
It is recommended to consider a positive response to vasoreactivity testing by a reduction in mPAP ≥10 mmHg to reach an absolute value of mPAP ≤40 mmHg with an increased or unchanged CO[c]	I	C
Inhaled nitric oxide, inhaled iloprost, or i.v. epoprostenol are recommended for performing vasoreactivity testing	I	C
Vasoreactivity testing, for identifying candidates for CCB therapy, is not recommended in patients with PAH other than I/H/DPAH and in PH groups 2, 3, 4, and 5	III	C

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Recommendations for diagnostic strategy		
Echocardiography		
Echocardiography is recommended as the first-line, non-invasive, diagnostic investigation in suspected PH	I	B
It is recommended to assign an echocardiographic probability of PH, based on an abnormal TRV and the presence of other echocardiographic signs suggestive of PH (see *Table 10*)	I	B
It is recommended to maintain the current threshold for TRV (>2.8 m/s) for echocardiographic probability of PH according to the updated haemodynamic definition	I	C
Imaging		
Ventilation/perfusion or perfusion lung scan is recommended in patients with unexplained PH to assess for CTEPH	I	C
CT pulmonary angiography is recommended in the work-up of patients with suspected CTEPH	I	C
Routine biochemistry, haematology, immunology, HIV testing, and thyroid function tests are recommended in all patients with PAH, to identify associated conditions	I	C
Abdominal ultrasound is recommended for the screening of portal hypertension	I	C
Other diagnostic tests		
Pulmonary function tests with DLCO are recommended in the initial evaluation of patients with PH	I	C
Open or thoracoscopic lung biopsy is not recommended in patients with PAH	III	C
Recommendations for screening and improved detection of pulmonary arterial hypertension and chronic thrombo-embolic pulmonary hypertension		
Systemic sclerosis		
In patients with SSc, an annual evaluation of the risk of having PAH is recommended	I	B
In adult patients with SSc with >3 years' disease duration, an FVC ≥40%, and a DLCO <60%, the DETECT algorithm is recommended to identify asymptomatic patients with PAH	I	B
In patients with SSc, where breathlessness remains unexplained following non-invasive assessment, RHC is recommended to exclude PAH	I	C
CTEPH/CTEPD		
In patients with persistent or new-onset dyspnoea or exercise limitation following PE, further diagnostic evaluation to assess for CTEPH/CTEPD is recommended	I	C
For symptomatic patients with mismatched perfusion lung defects beyond 3 months of anticoagulation for acute PE, referral to a PH/CTEPH centre is recommended after considering the results of echocardiography, BNP/NT-proBNP, and/or CPET	I	C
Other		
Counselling regarding the risk of PAH and annual screening are recommended in individuals who test positive for PAH-causing mutations and in first-degree relatives of patients with HPAH	I	B
In patients referred for liver transplantation, echocardiography is recommended as a screening test for PH	I	C
Recommendations for evaluating the disease severity and risk of death in patients with pulmonary arterial hypertension		
It is recommended to evaluate disease severity in patients with PAH with a panel of data derived from clinical assessment, exercise tests, biochemical markers, echocardiography, and haemodynamic evaluations	I	B
Achieving and maintaining a low-risk profile on optimized medical therapy is recommended as a treatment goal in patients with PAH	I	B
For risk stratification at the time of diagnosis, the use of a three-strata model (low, intermediate, and high risk) is recommended, taking into account all available data including haemodynamics	I	B
For risk stratification during follow-up, the use of a four-strata model (low, intermediate–low, intermediate–high, and high risk) based on WHO-FC, 6MWD, and BNP/NT-proBNP is recommended, with additional variables taken into account as necessary	I	B

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Recommendations for general measures and special circumstances		
General measures		
Supervised exercise training is recommended in patients with PAH under medical therapy	I	A
Psychosocial support is recommended in patients with PAH	I	C
Immunization of patients with PAH against SARS-CoV-2, influenza, and *Streptococcus pneumoniae* is recommended	I	C
Diuretic treatment is recommended in patients with PAH with signs of RV failure and fluid retention	I	C
Long-term oxygen therapy is recommended in patients with PAH whose arterial blood oxygen pressure is <8 kPa (60 mmHg)[d]	I	C
In the presence of iron-deficiency anaemia, correction of iron status is recommended in patients with PAH	I	C
The use of ACEis, ARBs, ARNIs, SGLT-2is, beta-blockers, or ivabradine is not recommended in patients with PAH unless required by comorbidities (i.e. high blood pressure, coronary artery disease, left HF, or arrhythmias)	III	C
Special circumstances		
In-flight oxygen administration is recommended for patients using oxygen or whose arterial blood oxygen pressure is <8 kPa (60 mmHg) at sea level	I	C
Recommendations for women of childbearing potential		
It is recommended that women of childbearing potential with PAH are counselled at the time of diagnosis about the risks and uncertainties associated with becoming pregnant; this should include advice against becoming pregnant, and referral for psychological support where needed	I	C
It is recommended to provide women of childbearing potential with PAH with clear contraceptive advice, considering the individual needs of the woman but recognizing that the implications of contraceptive failure are significant in PAH	I	C
It is recommended that women with PAH who consider pregnancy or who become pregnant receive prompt counselling in an experienced PH centre, to facilitate genetic counselling and shared decision-making, and to provide psychological support to the patients and their families where needed	I	C
For women with PAH having termination of pregnancy, it is recommended that this be performed in PH centres, with psychological support provided to the patients and their families	I	C
As teratogenic potential has been reported in pre-clinical models for endothelin receptor antagonists and riociguat, these drugs are not recommended during pregnancy	III	B
Recommendations for the treatment of vasoreactive patients with idiopathic, heritable, or drug-associated pulmonary arterial hypertension		
High doses of CCBs are recommended in patients with IPAH, HPAH, or DPAH who are responders to acute vasoreactivity testing	I	C
Close follow-up with complete reassessment after 3–4 months of therapy (including RHC) is recommended in patients with IPAH, HPAH, or DPAH treated with high doses of CCBs	I	C
Continuing high doses of CCBs is recommended in patients with IPAH, HPAH, or DPAH in WHO-FC I or II with marked haemodynamic improvement (mPAP <30 mmHg and PVR <4 WU)	I	C
Initiating PAH therapy is recommended in patients who remain in WHO-FC III or IV or those without marked haemodynamic improvement after high doses of CCBs	I	C
CCBs are not recommended in patients without a vasoreactivity study or non-responders, unless prescribed for other indications (e.g. Raynaud's phenomenon)	III	C
Recommendations for initial oral drug combination therapy for patients with idiopathic, heritable, or drug-associated pulmonary arterial hypertension without cardiopulmonary comorbidities[e]		
Initial combination therapy with ambrisentan and tadalafil is recommended	I	B
Initial combination therapy with macitentan and tadalafil is recommended	I	B
Initial combination therapy with macitentan, tadalafil, and selexipag is not recommended	III	B

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Recommendations for sequential drug combination therapy for patients with idiopathic, heritable, or drug-associated pulmonary arterial hypertension		
General recommendation for sequential combination therapy		
It is recommended to base treatment escalations on risk assessment and general treatment strategies (see treatment algorithm)	I	C
Evidence from studies with a composite morbidity/mortality endpoint as the primary outcome measure		
The addition of macitentan to PDE5is or oral/inhaled prostacyclin analogues is recommended to reduce the risk of morbidity/mortality events	I	B
The addition of selexipag to ERAs[f] and/or PDE5is is recommended to reduce the risk of morbidity/mortality events	I	B
The addition of oral treprostinil to ERA or PDE5i/riociguat monotherapy is recommended to reduce the risk of morbidity/mortality events	I	B
The addition of bosentan to sildenafil is not recommended to reduce the risk of morbidity/mortality events	III	B
Evidence from studies with change in 6MWD as the primary outcome measure		
The addition of sildenafil to epoprostenol is recommended to improve exercise capacity	I	B
Evidence from studies with safety of combination therapy as primary outcome measure		
Combining riociguat and PDE5is is not recommended[g]	III	B
Recommendations for intensive care management for pulmonary arterial hypertension		
When managing patients with right HF in the ICU, it is recommended to involve physicians with expertise, treat causative factors, and use supportive measures, including inotropes and vasopressors, fluid management, and PAH drugs, as appropriate	I	C
Recommendations for lung transplantation		
It is recommended that potentially eligible candidates are referred for LTx evaluation when they have an inadequate response to oral combination therapy, indicated by an intermediate–high or high risk or by a REVEAL risk score >7	I	C
It is recommended to list patients for LTx who present with a high risk of death or with a REVEAL risk score ≥10 despite receiving optimized medical therapy including s.c. or i.v. prostacyclin analogues	I	C
Recommendations for pulmonary arterial hypertension associated with drugs or toxins		
It is recommended to make a diagnosis of drug- or toxin-associated PAH in patients who had relevant exposure and in whom other causes of PH have been excluded	I	C
In patients with suspected drug- or toxin-associated PAH, it is recommended to immediately discontinue the causative agent whenever possible	I	C
Recommendations for pulmonary arterial hypertension associated with connective tissue disease		
In patients with PAH associated with CTD, treatment of the underlying condition according to current guidelines is recommended	I	A
In patients with PAH associated with CTD, the same treatment algorithm as for patients with IPAH is recommended	I	C
Recommendations for pulmonary arterial hypertension associated with human immunodeficiency virus infection		
In patients with PAH associated with HIV infection, antiretroviral treatment according to current guidelines is recommended	I	A
Recommendations for pulmonary arterial hypertension associated with portal hypertension		
Echocardiography is recommended in patients with liver disease or portal hypertension with signs or symptoms suggestive of PH, and as a screening tool in patients evaluated for liver transplantation or transjugular portosystemic shunt	I	C
It is recommended that patients with PAH associated with portal hypertension are referred to centres with expertise in managing both conditions	I	C
Drugs approved for PAH are not recommended for patients with portal hypertension and unclassified PH (i.e. elevated mPAP, high CO, and a normal PVR)	III	C

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Recommendations for shunt closure in patients with pulmonary-systemic flow ratio >1.5:1 based on calculated pulmonary vascular resistance		
In patients with an ASD, VSD, or PDA and a PVR <3 WU, shunt closure is recommended	I	C
In patients with an ASD and a PVR >5 WU despite PAH treatment, shunt closure is not recommended	III	C
Recommendations for pulmonary arterial hypertension associated with adult congenital heart disease		
Risk assessment		
Risk assessment is recommended for patients with persistent PAH after defect closure	I	C
Treatment		
Bosentan is recommended in symptomatic patients with Eisenmenger syndrome to improve exercise capacity	I	B
In women with Eisenmenger syndrome, pregnancy is not recommended	III	C
In patients with Eisenmenger syndrome, routine phlebotomy to lower elevated haematocrit is not recommended	III	C
Recommendations for pulmonary arterial hypertension with signs of venous/capillary involvement		
A combination of clinical and radiological findings, ABG, PFTs, and genetic testing is recommended to diagnose PAH with signs of venous and/or capillary involvement (PVOD/PCH)	I	A
Identification of biallelic *EIF2AK4* mutations is recommended to confirm a diagnosis of heritable PVOD/PCH	I	A
Referral of eligible patients with PVOD/PCH to a transplant centre for evaluation is recommended as soon as the diagnosis is established	I	C
Lung biopsy is not recommended to confirm a diagnosis of PVOD/PCH	III	C
Recommendations for paediatric pulmonary hypertension		
Children		
It is recommended to perform the diagnostic work-up, including RHC and acute vasoreactivity testing, and treat children with PH at centres with specific expertise in paediatric PH	I	C
In children with PH, a comprehensive work-up for confirming diagnosis and specific aetiology is recommended (similar to that in adults, but adapted for age)	I	C
For confirming PH diagnosis, RHC is recommended, preferably before initiating any PAH therapy	I	C
In children with IPAH/HPAH, acute vasoreactivity testing is recommended to detect those who may benefit from CCB therapy	I	C
It is recommended to similarly define a positive response to acute vasoreactivity testing in children and adults by a reduction in mPAP ≥10 mmHg to reach an absolute value of mPAP ≤40 mmHg, with an increased or unchanged CO	I	C
In children with PAH, a therapeutic strategy based on risk stratification and treatment response is recommended, extrapolated from that in adults, but adapted for age	I	C
It is recommended to monitor the treatment response in children with PAH by serially assessing a panel of data derived from clinical assessment, echocardiographic evaluation, biochemical markers, and exercise tolerance tests	I	C
Infants		
It is recommended to screen infants with bronchopulmonary dysplasia for PH	I	B
In infants with (or at risk of) bronchopulmonary dysplasia and PH, treating lung disease—including hypoxia, aspiration, and structural airway disease—and optimizing respiratory support is recommended before initiating PAH therapy	I	B
Recommendations for pulmonary hypertension associated with left heart disease		
In patients with LHD, optimizing treatment of the underlying condition is recommended before considering assessment of suspected PH	I	A
RHC is recommended for suspected PH in patients with LHD, if it aids management decisions	I	C

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RHC is recommended in patients with severe tricuspid regurgitation with or without LHD prior to surgical or interventional valve repair	I	C
For patients with LHD and suspected PH with features of a severe pre-capillary component and/or markers of RV dysfunction, referral to a PH centre for a complete diagnostic work-up is recommended	I	C
In patients with LHD and CpcPH with a severe pre-capillary component (e.g. PVR >5 WU), an individualized approach to treatment is recommended	I	C
When patients with PH and multiple risk factors for LHD, who have a normal PAWP at rest but an abnormal response to exercise or fluid challenge, are treated with PAH drugs, close monitoring is recommended	I	C
Drugs approved for PAH are not recommended in PH-LHD[h]	III	A
Recommendations for pulmonary hypertension associated with lung disease and/or hypoxia		
If PH is suspected in patients with lung disease, it is recommended that echocardiography[i] be performed and results interpreted in conjunction with ABG, PFTs including DLCO, and CT imaging	I	C
In patients with lung disease and suspected PH, it is recommended to optimize treatment of the underlying lung disease and, where indicated, hypoxaemia, sleep-disordered breathing, and/or alveolar hypoventilation	I	C
In patients with lung disease and suspected severe PH, or where there is uncertainty regarding the treatment of PH, referral to a PH centre is recommended[j]	I	C
In patients with lung disease and severe PH, an individualized approach to treatment is recommended	I	C
It is recommended to refer eligible patients with lung disease and PH for LTx evaluation	I	C
In patients with lung disease and suspected PH, RHC is recommended if the results are expected to aid management decisions	I	C
The use of ambrisentan is not recommended in patients with PH associated with IPF	III	B
The use of riociguat is not recommended in patients with PH associated with IIP	III	B
The use of PAH medication is not recommended in patients with lung disease and non-severe PH[k]	III	C
Recommendations for chronic thrombo-embolic pulmonary hypertension and chronic thrombo-embolic pulmonary disease without pulmonary hypertension		
CTEPH		
Lifelong, therapeutic doses of anticoagulation are recommended in all patients with CTEPH	I	C
Antiphospholipid syndrome testing is recommended in patients with CTEPH	I	C
In patients with CTEPH and antiphospholipid syndrome, anticoagulation with VKAs is recommended	I	C
It is recommended that all patients with CTEPH are reviewed by a CTEPH team for the assessment of multimodality management	I	C
PEA is recommended as the treatment of choice for patients with CTEPH and fibrotic obstructions within pulmonary arteries accessible by surgery	I	B
BPA is recommended in patients who are technically inoperable or have residual PH after PEA and distal obstructions amenable to BPA	I	B
Riociguat is recommended for symptomatic patients with inoperable CTEPH or persistent/recurrent PH after PEA	I	B
Long-term follow-up is recommended after PEA and BPA, as well as for patients with CTEPH established on medical therapy	I	C
Recommendations for pulmonary hypertension centres		
It is recommended that PH centres provide care by a multidisciplinary team (cardiologist, pneumologist, rheumatologist, nurse specialist, radiologist, psychological and social work support, appropriate on-call expertise)	I	C
It is recommended that PH centres have direct links and quick referral patterns to other services (such as genetic counselling, PEA/BPA, LTx, adult congenital heart disease service)	I	C

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It is recommended that PH centres maintain a patient registry	I	C
It is recommended that PH centres collaborate with patient associations	I	C

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6MWD, 6-minute walking distance; ABG, arterial blood gas analysis; ACEi, angiotensin-converting enzyme inhibitor; ARB, angiotensin receptor blocker; ARNI, angiotensin receptor–neprilysin inhibitor; ASD, atrial septal defect; BNP/NT-proBNP, brain natriuretic peptide/N-terminal pro-brain natriuretic peptide; BPA, balloon pulmonary angioplasty; CCB, calcium channel blocker; CpcPH, combined post- and pre-capillary pulmonary hypertension; CPET, cardiopulmonary exercise testing; CT, computed tomography; CTD, connective tissue disease; CTEPD, chronic thrombo-embolic pulmonary disease; CTEPH, chronic thrombo-embolic pulmonary hypertension; DLCO, Lung diffusion capacity for carbon monoxide; DPAH, drug-associated pulmonary arterial hypertension; ERA, endothelin receptor antagonist; FVC, forced vital capacity; HF, heart failure; HFpEF, heart failure with preserved ejection fraction; HIV, human immunodeficiency virus; HPAH, heritable pulmonary arterial hypertension; ICU, intensive care unit; I/H/DPAH, idiopathic, heritable, drug-associated pulmonary arterial hypertension; IIP, idiopathic interstitial pneumonia; ILD, interstitial lung disease; IPAH, idiopathic pulmonary arterial hypertension; IPF, idiopathic pulmonary fibrosis; i.v., intravenous; LHD, left heart disease; LTx, lung transplantation; mPAP, mean pulmonary arterial pressure; PAH, pulmonary arterial hypertension; PAWP, pulmonary arterial wedge pressure; PCH, pulmonary capillary haemangiomatosis; PDA, patent ductus arteriosus; PDE5i, phosphodiesterase 5 inhibitor; PE, pulmonary embolism; PEA, pulmonary endarterectomy; PFTs, pulmonary function tests; PH, pulmonary hypertension; PH-LHD, pulmonary hypertension associated with left heart disease; PVOD, pulmonary veno-occlusive disease; PVR, pulmonary vascular resistance; RHC, right heart catheterization; RV, right ventricular; s.c., subcutaneous; SGLT-2i, sodium–glucose cotransporter-2 inhibitor; SSc, systemic sclerosis; TRV, tricuspid regurgitation velocity; VKA, vitamin K antagonist; VSD, ventricular septal defect; WHO-FC, World Health Organization functional class; WU, Wood units.

[a]Class of recommendation.

[b]Level of evidence.

[c]Testing should also be performed in patients with a baseline mPAP ≤40 mmHg, in whom the same responder criteria apply.

[d]Measured on at least two occasions.

[e]Cardiopulmonary comorbidities are predominantly encountered in elderly patients and include risk factors for HFpEF such as obesity, diabetes, coronary heart disease, a history of hypertension, and/or a low DLCO.

[f]ERA used in the GRIPHON study were bosentan and ambrisentan.

[g]The PATENT plus study investigated the combination of sildenafil and riociguat; however, combining riociguat with any PDE5i is contraindicated.

[h]Safety concerns have been identified when ERAs are used in patients with HF (HFpEF and HFrEF, with or without PH) and when sildenafil is used in patients with persistent PH after correction of valvular heart disease.

[i]Assessments should ideally be made when the patient is clinically stable, as exacerbations can significantly raise pulmonary artery pressure.

[j]This recommendation does not apply to patients with end-stage lung disease who are not considered candidates for LTx.

[k]This does not include inhaled treprostinil, which may be considered in patients with PH associated with ILD irrespective of PH severity.

'What to do' and 'What not to do' messages developed with GRADE Evidence to Decision framework

GRADE				
Recommendations	**Quality of evidence**	**Strength of recommendation**	**Class[a]**	**Level[b]**
In patients with IPAH/HPAH/DPAH who present at low or intermediate risk of death, initial combination therapy with a PDE5i and an ERA is recommended	Low	Conditional	I	B
The use of PDE5i in patients with HFpEF and isolated post-capillary PH is not recommended	Low	Conditional	III	C
The use of PDE5i in patients with ILD and non-severe PH is not recommended	Very low	Conditional	III	C

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DPAH, drug-associated pulmonary arterial hypertension; ERA, endothelin receptor antagonist; HPAH, heritable pulmonary arterial hypertension; HFpEF, heart failure with preserved ejection fraction; ILD, interstitial lung disease; PDE5i, phosphodiesterase 5 inhibitor; PH, pulmonary hypertension.

[a]Class of recommendation.

[b]Level of evidence.

16. Quality indicators

Quality indicators (QIs) are tools that may be used to evaluate care quality, including structural, process, and outcomes of care.[847] They may also serve as a mechanism for enhancing adherence to guideline recommendations through associated quality-improvement initiatives and benchmarking of care providers.[848,849] As such, the role of QIs in improving care and outcomes for cardiovascular disease is increasingly being recognized by health care authorities, professional organizations, payers, and the public.[847]

The ESC understands the need for measuring and reporting quality and outcomes of cardiovascular care, and has established methods for developing the ESC QIs for the quantification of care and outcomes for cardiovascular diseases.[847] To date, the ESC has developed QI suites for a number of cardiovascular diseases[850–852] and embedded these in respective ESC Clinical Practice guidelines.[27,477,853,854] Furthermore, the ESC aims to integrate its QIs with clinical registries such as the EurObservational Research Programme and the European Unified Registries On Heart Care Evaluation and Randomized Trials (EuroHeart) project[855] to provide real-world data about the patterns and outcomes of care for cardiovascular disease across Europe.

In parallel with the writing of this Clinical Practice Guideline, a process has been initiated to develop QIs for patients with PH using the

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ESC methodology and through collaboration with domain experts and the Heart Failure Association of the ESC. Such QIs may be used for evaluating the quality of care for patients with PH, and enable important aspects of care delivery to be captured. These QIs, alongside their specifications and development process, will be published separately.

17. Supplementary data

Supplementary data is available at *European Heart Journal* online including key narrative question (1-8) and PICO questions (I-IV).

18. Data availability statement

No new data were generated or analysed in support of this research.

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20. Appendix

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